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Human resource allocation problem in the Industry 4.0: A reference framework

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ABSTRACT

The Human Resource Allocation Problem (HRAP) is a well-known mathematical modeling type where a set of tasks should be assigned to a group of operators. As the smart production system brought by Industry 4.0 (I4.0) requires aligning new technologies with the human perspective, existing HRAP models should be revised and adapted to this new context. Therefore, this paper aims to design a complete proposal to support the modeling, implementation, and dynamic use of the HRAP in I4.0 called HRAP-4.0. To do so, first, the I4.0 factors that affect the HRAP are identified to revise to what extent existing models in the literature address them. To bridge gaps, this paper proposes contributions in two areas: mathematical models and methodological aspects. First, new mathematical formulations of optimization strategies (objectives) and constraints are modeled to include the I4.0 factors in a modular and reusable way, generating HRAP-4.0 model building block categories. Then a reference framework to support the formulation of HRAP-4.0 adapted to specific I4.0 situations based on the categories above is presented. Last, an implementation methodology is proposed to integrate the framework into a real I4.0 context and use the resulting HRAP-4.0 models in a dynamic environment. The reference framework comprises two decision trees to support selecting the most suitable optimization strategies (objectives) and constraints categories for the HRAP-4.0 under study. It collects the existing model categories in the literature, plus the new ones related to I4.0 herein developed. They are all essential within the cyber-physical systems (CPS) frame, which is the base of I4.0. The proposed HRAP-4.0, thus, becomes the new original tool for assigning tasks to people in the modern production system.

1. Introduction

1.1. Scope and relevance

The Human Resource Allocation Problem (HRAP) is a variation of the original Allocation Problem (AP)(Kuhn, 1955), which consists of assigning limited resources to several tasks for optimizing one objective or more subject to the given resources' constraints. The importance of the AP relies on the fact that it has been the base for many other variations, e.g., production orders to production lines, vehicles to routes, products to production plants, raw materials to suppliers, among a very wide-open variety of situations (Pentico, 2007). More specifically, when allocated resources are persons, the problem is known as the HRAP (Bouajaja and Dridi, 2017). Therefore, the HRAP is one specific variation of the AP that maintains its original essentials of the relationship between tasks and persons with an additional complexity due to the human factor (Bouajaja and Dridi, 2017).

The HRAP plays a crucial role in industry because managing human resources implies controlling workloads. They are, by definition, sources of physical and mental stress, as well as conflicts among people because of their idle time, decisions on working shifts, persons per working shifts, tasks per working shift, and production capacity based on human resources availability. Besides, human resources management in industry and operational research is a constantly changing area due to newer manufacturing technologies and more possibilities to improve performance in operations.

Recently, such evolution has been affected by Industry 4.0 (I4.0) and its inherent tools. Industry 4.0 (Xu et al., 2018) was initially coined in 2011 in German Government project proposals for their High-Tech Strategy 2020. The Industry 4.0 paradigm implies digitizing the value chain and the interconnection of people, objects, and systems through real-time data exchange. The incorporation of artificial intelligence (AI)

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allows them to adapt to spontaneous changes in the environment independently and transform them into smart objects that enhance the creation of flexible self-controlled production systems (Hecklau et al., 2016). I4.0 involves accepted concepts and tools, such as AI, automation and robotics, the Internet of Things (IoT), big data, and cloud computing, coordinated in cyber-physical systems (CPS). The fourth industrial revolution is, in fact, the revolution of CPS. CPS are defined as "integrated systems engineered to combine computational control algorithms and physical components such as sensors and actuators, effectively using an embedded communication core" (Alwan et al., 2022). They run according to three fundamental factors:

- interoperability (Wegner, 1996); related to the ability of two software systems or more to cooperate;
- interconnectivity (Lima-Monteiro et al., 2016); for the system ability to share access to data warehouses and inputs (sensors and terminals);
- virtuality, related to CPS's ability to operate in a combined physical and virtual way.

I4.0 has brought about an enormous change in production systems, especially in flexibility in manufacturing, highly customized products, and shorter lead times. It is done by applying continuous real-time control to the service/production rate to deliver a quick response with maximum deployment (especially for digital environments) (Rennung et al., 2016). Production systems have three critical characteristics supported by the aforementioned CPS features in I4.0: traceability, flexibility, and resilience. Such elements are meant to accelerate the achievement of customer requirements.

Even when I4.0 tools might be considered replacements for human resources in performing activities, people still play a crucial role. Integrating workers into an I4.0 system with different skills, levels of education, and cultural backgrounds is a significant challenge. In this context, the so-called Human Cyber-Physical Systems (HCPS) are designed to facilitate cooperation between humans and machines (Villalba-Diez and Ordieres-Meré, 2021). Indeed CPS can be implemented to either substitute human input or collaborate with it, and HCPS are an example of applications that maintain accountable human input during a process (Flores et al., 2020). They play a significant role in the HRAP because they link the physical system, humans, and the cyber-physical space by integrating human labor into the I4.0 context. New production systems in I4.0 should be designed to enhance human capabilities. Indeed CPS support different human roles (Şahinel et al., 2021), such as: (1) data acquisition; (2) state inference; (3) state and system influencing; 4) actuation. Lu (2017) states that based on the IoT, CPS communicate and cooperate among them and with humans in real-time. This author expounds that CPS can increase productivity, enhance growth and improve workforce performance.

The I4.0 context needs a more technically specialized workforce to control the performance of the tasks that are either owned or supported by CPS in the workstation. In manufacturing, the collaborative workstation (Mateus et al., 2019) is an example of combining I4.0 tools with a highly skilled workforce. The collaborative workstation is an entirely feasible idea that can be extended to services. A self-sufficient workforce is required to facilitate flexibility, traceability, and resilience; and accelerate the performance of activities without supervision. It is an operational excellence principle (Rusev and Salonitis, 2016), the capacity to self-guide when performing activities, solving problems, and facilitating self-assessment. Lasi et al. (2014) call this feature in production systems decentralization. It bases its function on self-organization. When implementing I4.0 components, production systems are now commonly called smart production systems. They are defined as incorporating automatic learning, reconfigurability, human–machine collaboration, and comprehensive models to support the system, productivity, cost performance, and flexibility (Slotwinski and Tilove, 2007). In a decentralized schema, highly trained people, capable of responding to intelligent production system requirements from both qualification and aptitude terms, are known in this paper as the flexible workforce. Further on, it refers to it when talking to the workforce, labor, person, and operator. New methods and models are required to incorporate the online and offline massive amounts of data in decision making that put intelligent production systems to good use and enhance operators' roles and well-being. The integration of humans into smart manufacturing contexts will require new capabilities of operators. It also requires considering new decision support tools to improve the management and control of production and logistics systems and operators' well-being. As production and logistics systems need to align new technologies with the human perspective, existing HRAP models should be revised and adapted to this new context through the so-called HRAP-4.0.

1.2. Scientific contributions, and research questions

Mathematical programming is selected as the modeling technique in this research work because it is one of the most widely used methods to model the HRAP, which is later solved using exact, heuristic, or metaheuristics methods (Bouajaja and Dridi, 2017). However, despite the extensive literature on the topic, no study has revised HRAP models in the I4.0 context to adapt them to different scenarios. Therefore, this paper aims to bridge this gap by providing a complete proposal to model, implement, and use the HRAP in the I4.0 context (HRAP-4.0). Hence the contribution of this paper is manifold: 1) to analyze the impact of I4.0 on the HRAP by identifying the new aspects to be considered in this problem; 2) to define a unified nomenclature and formulate existing HRAP models in a modular and reusable way with building blocks categories; 3) to model the characteristics that derive from I4.0 (HRAP-4.0) with newer building blocks categories; 4) to provide a reference framework to support the proper selection of these categories to formulate HRAP-4.0 models adapted to each specific I4.0 situation; 5) to methodologically describe how to define, implement and use the generated HRAP-4.0 model to allocate human resources in a dynamic HCPS context. More specifically, this work aims to answer the following research questions (RQ):

- RQ1: What does I4.0 bring about modifying factors, and how could they affect the HRAP formulation?
- RQ2: To what extent have those factors been incorporated into existing HRAP models and what are the identified gaps?
- RQ3: How can a reference framework be defined to support the formulation of new HRAP-4.0 models for real situations?
- RQ4: Is it possible to formulate new HRAP models that put the I4.0 (HRAP-4.0) to good use to bridge any gaps detected in the literature?
- RQ5: How can these new HRAP-4.0 models be integrated into I4.0 technology (HCPS) to be used in a real environment?

1.3. Methodology

In order to address the above RQs, a research methodology was followed that adopts mathematical programming as the approach to model HRAP-4.0. It is assumed that the new I4.0 factors which affect the HRAP might require the modification of existing models to put new I4.0 technologies in practice. These modifications may imply the formulation of new models' indices, sets, parameters, decision variables, objectives, or constraints. It is also assumed that the resulting new mathematical formulations can be combined with already existing models to adapt the HRAP to each I4.0 situation. Therefore, when mathematically analyzing existing HRAP models and formulating new HRAP-4.0 ones, two principles were considered: modularity and reusability. Modularity because it is assumed that each model is integrated by different building blocks that represent specific objectives (optimization strategies) and constraints categories. Reusability because these previous building blocks can be combined with existing or new models to adapt the formulation of the HRAP to each specific I4.0 situation. So to combine existing models along with the new ones proposed in this paper, it was necessary to define a new homogeneous nomenclature.

Based on the above assumptions, the methodology followed to conduct this research is presented in Fig. 1, which coincides with the structure of the paper.

The first part of the paper serves as the theoretical basis to develop the contributions of this research. The theoretical foundation lies in the literature review (Section 2). In Section 2.1, the main I4.0 factors that affect the HRAP are identified and answer RQ1. These factors are the starting point to inductively derive the requirements and the necessary extension categories of the building blocks for HRAP-4.0. Section 2.2 offers the literature review conducted to identify the building blocks integrating current HRAP models and to what extent they cover the I4.0 factors identified in Section 2.1. This analysis answers RQ2 by identifying gaps in the literature (Section 2.3); it is the I4.0 factors that are not considered in existing HRAP models and need to be covered.

The second part of the paper looks to answer RQ3 and RQ4. It is done by describing the reference framework to support new HRAP-4.0 models. This framework is based on two elements: a) two decision trees that support the selection of the HRAP-4.0 building blocks adapted to the situation under study (Section 3); b) the mathematical modeling of the new building blocks for the HRAP-4.0 (Section 4) that cover the gaps detected in Section 2.3.

The third part (Section 5) describes the methodology proposed for applying the reference framework and using the resulting HRAP-4.0 model in a dynamic environment of a CPS. This section provides the answer to RQ5. The fourth part of the paper (Section 6) discusses the

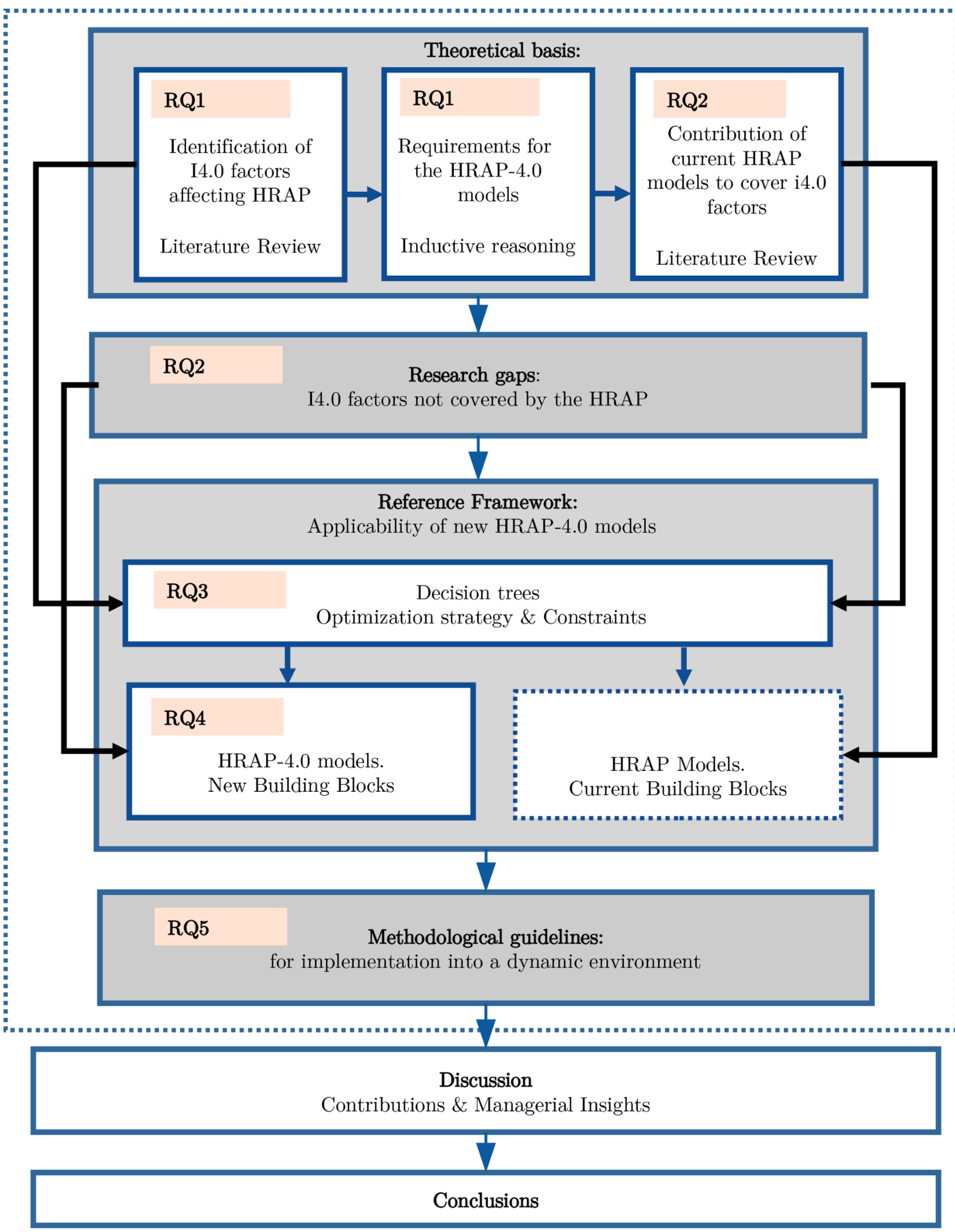


Fig. 1. Research methodology.

paper's contributions and its practical implications in the form of managerial insights. Finally, some conclusions, the limitations of this research, and future research lines are outlined (Section 7).

2. The HRAP literature review

The literature review is organized into two parts to support the HRAP-4.0 reference framework. First, in Section 2.1, the relevant implications of I4.0 in the HRAP are analyzed. This is done so that the main I4.0 factors that drive the extension of the HRAP are identified. These factors are inductively associated, based on existing research, with specific new requirements for the HRAP-4.0 building blocks (categories of optimization strategies and sets of constraints). The current HRAP building blocks in the literature are revised and analyzed in Section 2.2 to explore which I4.0 factors have not already been modeled. Combining both provides the existing gaps and requirements to extend HRAP models to HRAP-4.0 and its reference framework.

2.1. Implications of the I4.0 on the HRAP

2.1.1. HRAP modifying factors

The transformation of production systems deriving from I4.0 is relevant for the HRAP because of its role in human resources management (Barišić et al., 2019), mainly when the human-technology integration affects humans' performance (Barišić et al., 2019; Turulja and Bajgorić, 2016). This subsection seeks to answer RQ1 about how the factors brought about by I4.0 could affect the HRAP formulation. A factor is understood as a circumstance that potentially modifies or extends the HRAP due to I4.0. A brief review was conducted and refined on both concepts, I4.0, and human resources, to identify the main factors. They are inductively taken from the reviewed literature. In the following paragraphs, a short description is provided with representative

Table 1
HRAP-4.0 extension categories based on I4.0 factors.

			Optimization strategy categories		Constraints categories		Rolling horizon modeling	Reference framework	Methodological guidelines
I4.0 Factor	**Ref.**	**Requirements for the HRAP-4.0**	Workload balance and idle time	Occupational well-being	Time disruption	Recurrence			
Human–machine integration.	(Villalba-Diez and Ordieres-Meré, 2021; Lu, 2017; Slotwinski and Tilove, 2007; Barišić et al., 2019; Mateus et al., 2019; Brito et al., 2019; Romero et al., 2018; Ansari et al., 2018; Lorenz et al., 2015; Thoben et al., 2017; Zhong et al., 2017)	Human mental and physical well-being.	x	x				x	x
		Workload balance among persons.	x	x				x	x
		Assignation frecuency.				x	x	x	x
		Operator's real time measuring.	x		x		x	x	x
Human role in the CPS.	(Lu, 2017Slotwinski and Tilove, 2007Kolberg and Zühlke, 2015Thoben et al., 2017; Achillas et al., 2015Gregori et al., 2017)	Design CPS to support humans.	x	x				x	x
Process control.	(Lu, 2017; Bragança et al., 2019; Ansari et al., 2018; Lorenz et al., 2015; Thoben et al., 2017; Alcácer and Cruz-Machado, 2019)	Resilience to unexpected events.			x		x	x	x
		Continuity in task execution.			x		x	x	x
		Recurrence in task execution.			x	x	x	x	x
Information handling.	(Slotwinski and Tilove, 2007; Barišić et al., 2019; Barišić et al., 2019; Gregori et al., 2017; Alcácer and Cruz-Machado, 2019; Gehrke et al., 2015)	Analysis of human resource productivity.	x					x	x
		Improving learning curves.		x				x	x

references. The entire group of references associated with each corresponding factor is presented in the first column of Table 1. They all stress the need to update human resources management in I4.0 because of its effect on the production system.

Human–machine integration: I4.0 describes new paradigms for the integrated human–machine interaction. Both concepts are based on intelligent and interconnected cyber-physical production systems capable of controlling the process flow of industrial production and explaining the variability associated with value creation processes (Villalba-Diez and Ordieres-Meré, 2021). Authors like Mateus et al. (2019) and Brito et al. (2019) highlight its importance, especially for the collaborative workstation. The development of new technologies could impact operators' capabilities through CPS. This implies a highly trained and skilled workforce to perform multiple dynamic tasks in the production system. To increase productivity, performance, and workforce security and satisfaction, the main requirements to extend the HRAP deriving from this factor can be summarized as follows:

- Human mental and physical well-being: because of the flexibility requirements in an I4.0 context and operators' multidisciplinary profile, it is essential to focus on persons and their well-being in mental and physical aspects when assigning tasks (Brito et al., 2019; Romero et al., 2018). Optimization strategies should include occupational satisfaction and security (occupational well-being) while allocating tasks.
- Workload balance among persons: this requirement is highlighted as an analysis trend in I4.0 (Brito et al., 2019; Romero et al., 2018) to avoid overburden. The social interaction between humans will require more care in the workload balance. Expectations, qualifications, and the type of relationship with different machines might generate significant imbalances. The workload itself is a factor in controlling occupational motivation. Labor productivity can be linked with idle time. By decreasing idle time and workload deviations, the average productivity rate goes up.
- Assignation frequency: Facilitating interoperability and interconnectivity requires integrated people (Lu, 2017) whose performance will be affected by the turbulence, randomness, and technical nature of the CPS. It will bring new tasks and outdating others. The new ones can imply deeper cognitive load requirements and higher execution frequency to interact with the CPS.
- Operators' real-time measuring: there is consensus (Villalba-Diez and Ordieres-Meré, 2021; Lu, 2017; Barišić et al., 2019; Mateus et al., 2019; Brito et al., 2019; Romero et al., 2018) that CPS impact humans' performance. Even when automation does not directly imply its enhancement, humans limit the cognitive load that affects their work (Villalba-Diez and Ordieres-Meré, 2021). Real-time measuring provides operators with more effective feedback. It relates to their productivity rate and the definition of training programs. It also allows anticipating whether a task needs to be stopped or not (disruption) and if it is necessary to execute it more than once on a planning horizon (recurrence).

Human role in the CPS: Kolberg and Zühlke (2015) state that humans take a central position in I4.0; supported by innovative information and communication technologies, they become smart operators who supervise and control ongoing activities themselves. Therefore, operators should be considered part of CPS, and CPS should be capable of facilitating the planning of their tasks while enabling the system's performance and productivity improvement and satisfaction at work. The main requirement that derives from this factor is summarized as follows:

- Design CPS to support humans: humans will be integrated into CPS to make production more cost- and time-efficient with highly qualified products (Lu, 2017). They should support one another while prioritizing protecting people. A new version of the HRAP should seek to facilitate this fact.

Process control: Lu (2017) states that CPS communicate and cooperate among themselves and with humans in real-time based on the IoT. CPS also allow online process control to detect task interruptions. Indeed CPS should support real-time assignments in service to facilitate flexibility to react to sudden changes and, therefore, toward resilience. The specific requirements deriving from this factor are:

- Resilience to unexpected events: in this case, the HRAP should react to events that could provoke unfinished tasks or interruptions in their execution. Humans will still be better able to reason and make the decisions that autonomous systems cannot yet replace (Bragança et al., 2019). This will be important when triggering events occur that generate disruptions in CPS operations, and humans should react quickly and effectively. A re-assignation or reconfiguration might be necessary to maintain the system's steady state.
- Continuity in task execution: as an extension of the reaction to unexpected events, task execution can require additional tools to ensure continuity. That is, avoiding or allowing interruptions depending on each specific case.
- Recurrence in task execution: Executing the tasks on a planning horizon can be highly recurrent. Repetitive tasks will be shared between automated solutions and humans in the collaborative workstation (Bragança et al., 2019). Accordingly, HRAP-4.0 should ensure time windows between successive executions for the same task, e.g., manual data inputs for CPS.

Information handling: this factor stems from the fact that managing a large amount of information and data, combined with interacting with complex machines and systems, will be the essential element of future work tasks (Gehrke et al., 2015). The generation and use of data by CPS should ensure massive availability for online or offline use. The derived requirements are:

- Analysis of human resource productivity: human resources management should increase productivity, accelerate response times and improve decision-making processes (Barišić et al., 2019; Barišić et al., 2019). The pertinence and availability of more precise and descriptive data update human productivity for more efficient task assignations.
- Improving learning curves: transforming learning by improving the performance and quality of human resources is also the consequence of digital transformation (Barišić et al., 2019). Tracking productivity improves learning curves with the correct definition of training programs.

2.1.2. Extensions for HRAP-4.0 building blocks

The new requirements for the HRAP driven by the previously analyzed I4.0 factors should be translated into suitable modeling extensions for the new HRAP-4.0 models. It is also necessary to provide guidelines about developing and executing HRAP-4.0 models from a methodological point of view. This work proposes extensions of the building blocks in the HRAP and some methodological proposals:

- Optimization strategies based on workload balance and idle time focused on protecting persons: the integration of CPS and HRAP should seek to prioritize persons. Operators' real-time measuring can detect workload imbalances.
- Optimization strategies based on occupational well-being: it incorporates into the HRAP-4.0 people's needs when assigning tasks within the CPS framework. These needs should reflect satisfaction with task execution.
- Time disruption constraints: this category is considered to tackle the need that HRAP-4.0 will deal with more variety and technical

requirements during tasks. A key characteristic in executing them in the CPS context is dealing with disruptions. A disruption involves interrupting a task temporarily until it can be finished.

- Recurrence constraints: this category refers to the need to execute one task several times on a planning horizon, depending on CPS requirements.
- Rolling horizon: to adapt the assignation of human resources to environment changes, new HRAP-4.0 model executions with updated information should be done every replanning period by considering the current progress status of tasks and their present allocated persons. Besides, given the online information provided by CPS, events can be detected that require a more frequent re-assignation of tasks to persons.
- Reference framework: used to support selecting the most suitable optimization strategies and constraints for each I4.0 situation.
- Methodological guidelines for implementation: this category refers to the need to execute HRAP-4.0 while systematically interacting with CPS in a dynamic environment. This methodology will use new HRAP-4.0 models as input.

The relationship of the proposed extension categories for HRAP-4.0 building blocks and the corresponding requirements driven by the I4.0 factors analyzed in Section 2.1.1 are shown in Table 1. A category is understood as a group of similar optimization strategies and constraints approach given its purpose. Table 1 aims to indicate how the I4.0 modifying factors and their corresponding requirements for HRAP-4.0 are associated with the proposals of this work. It is worth noting that only the closest relations between the I4.0 requirements and the categories of this proposal are analyzed.

Therefore, mental and physical well-being and workload balance among persons are required to reflect the human–machine integration factor. Two optimization strategies address these requirements: workload balance and occupational well-being. By balancing workload, human mental and physical well-being is positively impacted simultaneously. Assignation frequency is directly linked with the recurrence constraint category. The times that the same task is executed on a planning horizon are interpreted as recurrence. Finally, operators' real-time measuring can be directly used to detect workload imbalances, tasks being executed, finished, or interrupted. This requirement is linked with the workload balance optimization strategy and time disruption constraints. It is essential to clarify that all the HRAP-4.0 requirements related to the constraint categories can be directly linked with the rolling horizon planning because both time disruption and recurrence for the same task can be managed with different iterations of the same HRAP-4.0 model.

As for the human role in CPS, the derivated requirement of designing CPS to support humans is linked with the workload balance and occupational well-being. Given that both aspects intend to prioritize persons, it is consequent with supporting humans. In the process control factor, note that the primary relation of requirements is linked with the constraint categories. By monitoring the process using CPS, the execution of the tasks required by it becomes crucial for measuring performance. Such execution is subject to the task's required time and how it advances. Hence, it could be susceptible to disruptions and recurrence.

The information handling factor is associated with the workload balance through the human resource productivity factor. It is considered that a person's workload is a determining factor for their performance on the shift. So information about assigned tasks and the total workload is required to measure their results correctly. This requirement could also be related to the disruption constraints, but it is considered that workload is dominant. The improving learning curves requirement is linked to the occupational well-being optimization strategy because the better the person feels on the task, the easier it is for them to learn and explore better ways of doing flexible activities.

Finally, it is important to stress that a reference framework is proposed to support modeling the above optimization strategies and constraint categories. For this reason, the reference framework is linked with all the HRAP-4.0 requirements. Once HRAP-4.0 is modeled, it needs to be implemented and used dynamically. To help with this task, a methodological guideline is proposed to cover all requirements. The categories are defined in Table 1 so that they are general enough to cover all the identified factor's requirements. These relations are the basis for generalizing to HRAP-4.0. More positive relations in Table 1 imply a greater need to overcome the drawback.

2.2. Current HRAP building blocks

This section aims to address RQ2, mapping to what extent current HRAP models have incorporated the I4.0 factors presented in Section 2.1. For this purpose, different phases were carried out: 1) the categorization of existing optimization strategies and constraints in the HRAP models of the revised literature; 2) the definition of a unified notation; 3) the formulation of the identified optimization strategies and constraint categories in a modular and reusable way formulating building blocks based on the previously unified notation; 4) the verification that the HRAP-4.0 extension categories based on I4.0 factors have not been formulated in previous research.

First, two previous literature reviews were selected as a basis for our review (Pentico, 2007; Bouajaja and Dridi, 2017). Pentico (2007) extensively reviews the development of the original AP, while Bouajaja and Dridi (2017) offer a specific review of HRAP applications and their different variations based on the optimization objectives and sets of constraints. This section recalls and updates the building blocks exposed by Pentico (2007) and Bouajaja and Dridi (2017). Second, to conduct the review, relevant scientific databases (i.e., Scopus, Web of Knowledge, Elsevier, Scholar Google, among others) were consulted using the following keywords as they are typically used to refer to a wide variety of HRAP versions:

- Human resource allocation problem.
- Allocation problem.
- Assignation problem.
- Task/job assignation.
- Time-lapse allocation problem.

Third, the found papers were refined to those of HRAP related and not related to I4.0; in all, 44 were selected. One-third of the papers were published in the last four years, as shown in Table 2.

Fourth, by characterizing the HRAP building blocks, the terminology shown in Table 3 was defined. All the notations in Table 3 can be expressed multi-dimensional by adding indices, such as index k for other resources or index t of time.

Fifth, the existing optimization strategies and constraints in Sections 2.2.1 and 2.2.2, respectively, are categorized inductively and grouped according to the similarities of their purpose. Such categories are considered those of the current HRAP building blocks. It is important to stress that none of these categories and their building blocks match neither the I4.0 factors presented in Section 2.1.1 nor the HRAP-4.0 requirements of Section 2.1.2. They point out the lacking aspects of the HRAP to be extended to HRAP-4.0. The purpose of this work is to extend those categories to new HRAP-4.0 formulations. These new

Table 2
Year of publication.

Year	References	%
2015 to 2019	13	29.5
2011 to 2015	3	6.8
2006 to 2010	2	4.6
2000 to 2005	4	9.1
before 2000	22	50
Total	44	100

Table 3
Terminology to formulate revised HRAP models.

Model section	Name	Description
Indices	j	Tasks (jobs) to be assigned $j \in \{1, \dots, J\}$.
	p	Persons $p \in \{1, \dots, P\}$.
Parameters	$\mathbf{b}$	Total amount of an available resource.
	$\mathbf{c}_j^p$	Cost of assigning task j to person p.
	$\mathbf{d}^p$	Number of tasks that should be executed by each person p.
	$\mathbf{p}_j^p$	Profit of assigning task j to person p.
	$\mathbf{q}_j^p$	Binary parameter with a value of 1 if person p can perform task j.
	$\mathbf{r}_j^p$	Amount of a resource required by person p to perform task j.
	$\mathbf{t}_j^p$	Time required by person p to perform task j.
Decision variables	$\mathbf{X}_j^p$	Binary variable with a value of 1 if task j is assigned to person p.

models are integrated with the revised ones in Table 1 through a reference framework for HRAP-4.0 found in Section 3). The new model extensions are presented in detail in Section 4.

2.2.1. HRAP optimization strategies

The synthesis of the literature review results is shown in Table 4. The information in Table 4 is organized and complemented from that provided in Pentico (2007) and Bouajaja and Dridi (2017) in six main categories for the optimization strategies (objective functions): i) profit; ii) cost; iii) execution time; iv) ratios; v) permutations; vi) persons' qualifications. In each category, different alternatives exist that give rise to distinct building blocks. The percentages for each building block in Table 4 do not add up to 100%, as one same reference can have more than one optimization strategy. As discussed in Section 2.3, none of the current building block categories covers the I4.0 requirements shown in Table 1.

2.2.2. HRAP sets of constraints

Constraints information is also reorganized based on Pentico (2007) and Bouajaja and Dridi (2017) into three main categories: i) task per person, which are the constraints controlling the number of tasks assigned to each person; ii) persons per task, for the constraints acting oppositely, controlling the number of persons assigned to every task; iii) balance constraints, intended to limit the use of resources that might be needed to implement specific optimization strategies; for example, persons' qualifications of execution time. Each category can have more than one constraining purpose depending on the modeling situation. The results are shown in Table 5. Like the optimization strategies, none of the identified categories covers the I4.0 requirements presented in Table 1, as discussed in Section 2.3.

2.3. Discussion: From HRAP to HRAP-4.0

As regards optimization strategies, based on the references revised in Table 4, it can be stated that the minimization of the total cost is the most specific objective category and is addressed in 28.6% of the references. Nonetheless, the maximization of profit, persons' qualifications, and execution time are other similarly applied optimization strategies.

Permutation and ratios (seen as productivity indicators) are the least used strategies with only 4.8% of the references. Apart from cost and profit, the other strategies are not consistently studied. For example, the main focus is minimizing the last task's finishing execution time. There is no mention of other cases, such as the workload balance or human well-being, and even less so when combined with different optimization strategies like qualification, productivity ratios, and physical impact, identified as requirements derived from the I4.0 factors.

Although the specific case of workload balance has been studied in other situations, like the line balancing problem (LBP) (Schwerdfeger and Walter, 2018; Potts and Whitehead, 2001), it solves the assignation of a list of tasks to a list of workstations. A workstation is not necessarily given to only one person in production systems. Consequently, a workload balance for the person is not guaranteed. None of the existing workload balance methods focuses on compensation for persons. Classic models strike balances by looking for a predefined average. None considers the person's availability or the distance differences of workload without a predefined expected average. These are essential factors to consider in HRAP-4.0.

From the results in Table 5, it is clear that constraints link the task with the person. The HRAP constraint building blocks focus on limiting persons per task and tasks per person based on the number of observed references. However, only one work deals with persons' qualifications as an underresearched category that could be improved to be linked with I4.0 factors. Operators in I4.0 will face more variability in qualifications to work in CPS. Qualification can also allow combinations with newer optimization strategies, such as time balance and occupational well-being. In I4.0, it does not suffice to link the task with the person; the task itself can have specific characteristics during its execution time, like no interruptions or being executed more than once on that planning horizon. Such features are not reflected in the literature.

Moreover, no set of constraints considers executing the HRAP in a dynamic rolling horizon schema. Finally, having simultaneous pending and ongoing tasks is not adequately covered in current HRAP building blocks. As mentioned in Section 2.1.2, these last aspects are relevant in CPS (see Table 1).

In summary, the performed analysis of the existing building blocks in the literature for optimization strategies and constraints categories (Table 4 and Table 5) shows that the I4.0 requirements and categories in Table 1 are not addressed. Therefore, extending these current building block categories is necessary to adapt the HRAP formulation to the I4.0 context. The latter is also helpful for building a reference framework to guide users to select suitable building blocks for a specific I4.0 situation. To do so, this work identifies the following directions that are addressed in the sections below:

Optimization strategies based on workload balance and idle time:

- HRAP-4.0 should include additional components to seek a workload balance and idle time that focus on the person (not in the workstation), based on availability and abilities. The workload balance provides equivalent workloads to everyone if they have similar working shifts. Otherwise, it should minimize their idle time.

Optimization strategies based on occupational well-being:

- The CPS design should prioritize human well-being and production system productivity while facilitating flexibility, traceability, and resilience. Well-being can focus primarily on personal satisfaction and the physical risk of labor. The workload balance can also support human well-being.

Time disruption constraints:

- Implementing HRAP-4.0 into a rolling horizon schema is necessary to integrate it into a CPS properly. Certain events can also provoke the interruption of ongoing tasks that should be completed later. Both facts consequently bring out the need to consider possible time disruptions during task execution.

Recurrence constraints:

Table 4
Optimization strategies in the HRAP.

Optimization strategy category	Purpose	Objective function	References	(%)
Profit	Maximize the total profit associated with the assignation of tasks to the group of people.	$\sum_p\sum_j \mathbf{p}_j^p \mathbf{X}_j^p$	(Chang and Ho, 1998; Caron et al., 1999; Tadic and Marasovic, 2018; Nembhard and Bentefouet, 2015; Li et al., 2012; Aviso et al., 2017)	14.3
Cost	Minimize the total cost of assigning tasks to the group of people.	$\sum_p\sum_j \mathbf{c}_j^p \mathbf{X}_j^p$	(Caron et al., 1999; Dell'Amico and Martello, 1997; Prins, 1994; Kennington and Wang, 1992; Volgenant, 1996; Campbell and Diaby, 2002; Cattrysse and Van Wassenhove, 1992; Arias et al., 2017a; Arias et al., 2017b; Toroslu and Arslanoglu, 2007; Drezner, 2003; Tsui and Chang, 1992)	28.6
	Minimize the maximum of the assignment costs.	$\max_{j,p}\left\{\mathbf{c}_j^p \mathbf{X}_j^p\right\} \wedge \max_{j,p}\left\{\mathbf{c}_j^p \vert \mathbf{X}_j^p = 1\right\}$	(Ravindran and Ramaswami, 1977)	2.4
	Minimize the maximum of the sum of the costs for a person.	$\max_j \sum_p \mathbf{c}_j^p \mathbf{X}_j^p$	(Ravindran and Ramaswami, 1977; Arora and Puri, 1998; Fieldsend, 2017)	7.1
	Minimize the difference between the maximum and minimum assignment costs; m and n are the total number of tasks and persons respectively.	$\max_{j,p}\left\{\mathbf{c}_j^p \vert \mathbf{X}_j^p = 1\right\} - \min_{j,p}\left\{\mathbf{c}_j^p \vert \mathbf{X}_j^p = 1\right\}$	(Duin and Volgenant, 1991; Manavizadeh et al., 2013)	4.8
	Minimize the difference between the maximum and average assignment costs.	$\sum_{j,p}\left(\max_{m,n}\{\mathbf{c}_m^n \mathbf{X}_m^n\} - \mathbf{c}_j^p\right)\mathbf{X}_j^p$, if $m = n$, $\min\{m,n\}\cdot\left(\max_{m,n}\{\mathbf{c}_m^n \mathbf{X}_m^n\}\right) - \sum_{j,p} \mathbf{c}_j^p \mathbf{X}_j^p$, if $m \neq n$.	(Duin and Volgenant, 1991)	2.4
	Minimize the combined cost of the assignments plus the supervisory costs. F is the "supervisory" cost per period.	$\sum_{j,p} \mathbf{c}_j^p \mathbf{X}_j^p + F \cdot \max\left\{\mathbf{t}_j^p \vert \mathbf{X}_j^p = 1\right\}$	(Geetha and Nair, 1993; Scarelli and Narula, 2002; Ai, 2017)	7.1
	Minimize the total cost of assignments of tasks to a group of persons and machines simultaneously; k is the index of machine.	$\sum_{j,p,k} \mathbf{c}_{jk}^p \mathbf{X}_{jk}^p$	(Laguna et al., 1995; Gilbert and Hofstra, 1988; Gilbert and Hofstra, 1987; Al Khatib and Noppen, 2017)	9.5
Execution time	Minimize the time at which the last task is completed if all of them are start simultaneously in all the machines assigned; k is the index of machine.	$\max_{j,p,k}\left\{\mathbf{t}_{jk}^p \mathbf{X}_{jk}^p\right\}$	(Geetha and Vartak, 1994; Punnen and Aneja, 1993; Vartak and Geetha, 1990; Song et al., 2006)	9.5
Ratio	Minimize or maximize the ratio of two other objective expressions (productivity measure).	$\sum_{j,p} \frac{\mathbf{c}_j^p}{\mathbf{t}_j^p} \mathbf{X}_j^p$	(Shigeno et al., 1995; Volgenant, 2004)	4.8
Permutation	Case 1: minimize the time at which the last set of tasks is finished if all sets start simultaneously; the completion time is the sum of its tasks. Case 2: minimize the sum of the maximum times across the different sets that are to be done in sequence, but the tasks within each set may be done simultaneously. The index i represents the categories of tasks, and $C_{P(j)}$ is the permutation of tasks in category i.	$\text{Case 1}: \text{Min}_P \max_i \sum_{j \in S_i} C_{P(j),j}$ $\text{Case 2}: \text{Min}_P \sum_i \max_{j \in S_i}\{C_{P(j),j}\}$	(Punnen and Aneja, 1993; Aneja and Punnen, 1999)	4.8
Person qualification	Minimize the maximum number of persons assigned as they are qualified to perform the assigned tasks.	$\max_p \sum_j \mathbf{q}_j^p \mathbf{X}_j^p$	(Chang and Ho, 1998; Tadic and Marasovic, 2018; Li et al., 2012)	7.1
	From among all optimal solutions to the cardinality β-AP, identify that which maximizes the minimum weight of any assignment. A_j is the set of tasks that each person p is able to perform.	$\text{Max}_j \min_{p \in A_j}\left\{\mathbf{c}_j^p \mathbf{X}_j^p\right\}$ or $\text{Max}_j \min_p\left\{\mathbf{c}_j^p \mathbf{q}_j^p \mathbf{X}_j^p\right\}$	(Chang and Ho, 1998; Toroslu, 2003; Toroslu and Arslanoglu, 2007)	7.1

- When planning assignation over more than one period of time, i.e., on a planning horizon, recurrence in task executing is a realistic possibility. It is not contemplated in the HRAP, but it should be for HRAP-4.0, especially in the context of a CPS and a collaborative workstation.

Rolling horizon:

- The I4.0 context is characterized by its dynamism due to access to online data. HRAP-4.0 should contemplate updating the assignation of tasks by either period or event. This allows HRAP-4.0 to be

Table 5
Sets of constraint in the HRAP.

Constraint category	Purpose	Description	Constraint	References	(%)
Tasks per person	At most one task per person	Each task can be assigned only to one person or left without assignation.	$\sum_p X_j^p \leqslant 1, \forall j$	(Caron et al., 1999; Dell'Amico and Martello, 1997; Prins, 1994; Gilbert and Hofstra, 1988; Gilbert and Hofstra, 1987; Vartak and Geetha, 1990; Arias et al., 2017a; Arias et al., 2017b; Lira et al., 2017; Toroslu, 2003; Toroslu and Arslanoglu, 2007; Nembhard and Bentefouet, 2015; Li et al., 2012)	31.0
	One task per serveral persons	Each task can be shared and performed by more than one person, but at least one.	$\sum_p X_j^p \geqslant 1, \forall j$	(Chang and Ho, 1998; Campbell and Diaby, 2002; Cattrysse and Van Wassenhove, 1992; Arora and Puri, 1998; Duin and Volgenant, 1991; Gilbert and Hofstra, 1988; Arias et al., 2017a; Arias et al., 2017b; Lira et al., 2017; Toroslu, 2003; Toroslu and Arslanoglu, 2007; Li et al., 2012; Drezner, 2003; Shigeno et al., 1995; Tsui and Chang, 1992)	35.7
	One task, one person	Each task must be assigned to one person.	$\sum_p X_j^p = 1, \forall j$	(Kennington and Wang, 1992; Volgenant, 1996; Ravindran and Ramaswami, 1977; Duin and Volgenant, 1991; Scarelli and Narula, 2002; Gilbert and Hofstra, 1988; Geetha and Vartak, 1994; Punnen and Aneja, 1993; Fieldsend, 2017; Drezner, 2003; Shigeno et al., 1995; Tsui and Chang, 1992)	28.6
	One task, a group of persons	Each task must be assigned to a group of persons of a pre-defined size.	$\sum_p X_j^p = \mathbf{d}_j, \forall j$	(Ai, 2017)	2.4
	Tasks per qualified persons	The tasks must be assigned to qualified persons according to the four later category rules.	$\sum_p \mathbf{q}_j^p X_j^p \leqslant 1, \forall j$ $\sum_p \mathbf{q}_j^p X_j^p \geqslant 1, \forall j$ $\sum_p \mathbf{q}_j^p X_j^p = 1, \forall j$ $\sum_p \mathbf{q}_j^p X_j^p = \mathbf{d}_j, \forall j$	(Manavizadeh et al., 2013; Jin et al., 2017)	4.8
Persons per task	At most one person per task	Each person can be assigned only to one task or left with no assignation.	$\sum_j X_j^p \leqslant 1, \forall p$	(Caron et al., 1999; Dell'Amico and Martello, 1997; Prins, 1994; Vartak and Geetha, 1990; Lira et al., 2017; Toroslu, 2003; Li et al., 2012)	16.7
	One person for serveral tasks	Each person can be assigned with one task or more.	$\sum_j X_j^p \geqslant 1, \forall p$	(Lira et al., 2017; Toroslu, 2003; Li et al., 2012)	7.1
	One person one task	Each person must be assigned to one task.	$\sum_j X_j^p = 1, \forall p$	(Caron et al., 1999; Campbell and Diaby, 2002; Ravindran and Ramaswami, 1977; Duin and Volgenant, 1991; Geetha and Nair, 1993; Scarelli and Narula, 2002; Gilbert and Hofstra, 1987; Geetha and Vartak, 1994; Punnen and Aneja, 1993; Aneja and Punnen, 1999; Luo et al., 2017; Nembhard and Bentefouet, 2015; Shigeno et al., 1995; Volgenant, 2004)	33.3
	One person, a group of tasks	Each task in a group must be assigned and all the group of tasks assigned to the same person.	$\sum_j X_j^p = \mathbf{d}^p, \forall p$	(Kennington and Wang, 1992; Volgenant, 1996)	4.8
	Qualified persons per task	The qualified persons must be assigned to the tasks according to the four later category rules.	$\sum_j \mathbf{q}_j^p X_j^p \leqslant 1, \forall p$ $\sum_j \mathbf{q}_j^p X_j^p \geqslant 1, \forall p$ $\sum_j \mathbf{q}_j^p X_j^p = 1, \forall p$ $\sum_j \mathbf{q}_j^p X_j^p = \mathbf{d}^p, \forall p$	(Manavizadeh et al., 2013)	2.4
Balance constraints	Amount of tasks balance	Only m quantity of tasks can be assigned, where m is less than the total quantity of tasks and persons.	$\sum_{j,p} X_j^p = m$	(Aviso et al., 2017)	2.4
	Resource balance	The assignation should be done while contemplating a resource limitation.	$\sum_{j,p} \mathbf{r}_j^p X_j^p \leqslant \mathbf{b}$	(Cattrysse and Van Wassenhove, 1992; Laguna et al., 1995; Tadic and Marasovic, 2018; Lili, 2017)	9.5

reactive and resilient to events that can endanger the correct performance of assigned tasks.

Reference framework:

- A reference framework that provides a complete overview of existing HRAP building blocks with the new ones deriving from HRAP-4.0 is necessary. Besides, this reference framework will support the formulation of the HRAP-4.0 models adapted to the specific I4.0 situation.

Methodological guidelines for implementation:

- The new production system requires interaction between CPS and labor. It should happen in a decentralized execution-based schema using a flexible workforce.
- None of the reviewed references adequately addresses integrating the HRAP into the I4.0 context, nor provides any direction as to how to do this procedurally. Therefore, directions on integrating them into a CPS should be provided.

3. Reference framework for the HRAP-4.0

This section aims to answer RQ3. It presents a general reference framework to support the development of HRAP-4.0 models. It results from the extension requirements for the building block categories of the optimization strategies and sets of constraints identified in Section 2.1.2 and the current building block categories presented and analyzed in Section 2.2. The framework comprises two decision trees (Figs. 2 and 3): the first supports the modeler when selecting and formulating the optimization strategy (objective function); the second is for the constraints. Each decision tree integrates the corresponding categories defined in the literature review and those identified for I4.0 requirements. As observed in the symbology, each decision branch ends in a green, blue, or red circle. The green circle means that the corresponding strategy/constraints exist in the previous literature by specifying the building block categories of Table 4 or Table 5. The blue circle stresses that novel optimization strategies/constraints are proposed in this paper. These new HRAP-4.0 formulations are described in detail in Section 4. Finally, the red circle points out the gaps detected in the literature review that are not covered by this paper and, therefore, generates future research lines.

The decision trees are presented to guide modelers by developing the HRAP-4.0 model that better fits the assignation situation under study. Modelers can decide which equations compose the final model based on the chosen building block categories.

3.1. Optimization strategies of HRAP-4.0.

All the building blocks categories for the optimization strategies are included in the decision tree presented in Fig. 2. The framework discriminates among all categories, i.e., profit, cost, workload balance, human well-being, persons' qualifications, productivity ratios. The decision tree guides modelers through the specific route depending on the assignation situation. A multi-objective approach can also be applied if necessary, especially when conflicting strategies (opposite trends) are relevant. Additionally, it considers that the computational efficiency in the resolution process worsens for more significant instances of execution and multi-objective optimization.

3.2. Sets of constraints of HRAP-4.0.

The decision tree for the sets of constraints is presented in Fig. 3. For this case, the tree directly highlights that the current categories of tasks per person, persons per task, and balance constraints are mandatory as a

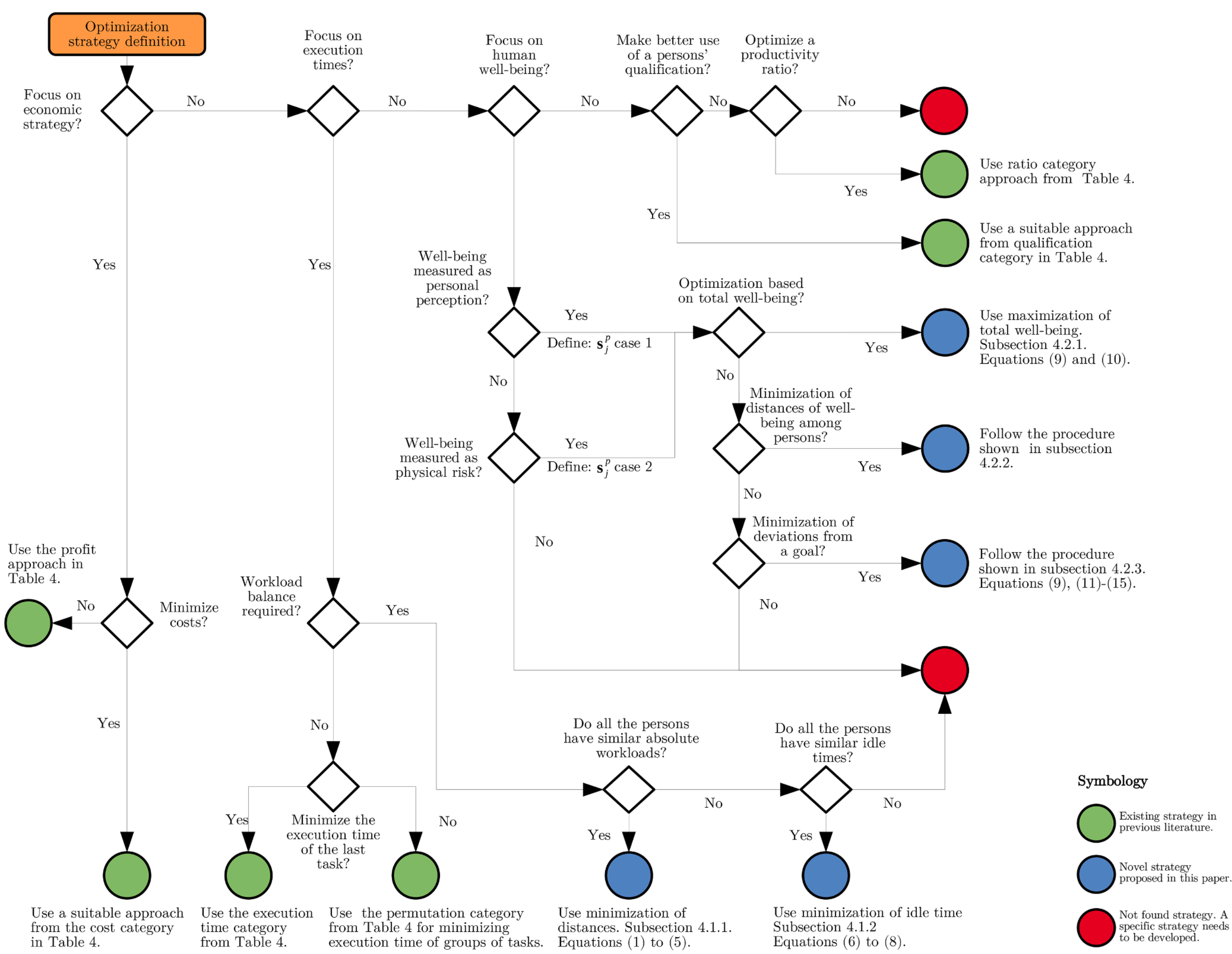


Fig. 2. Decision tree: optimization strategy.

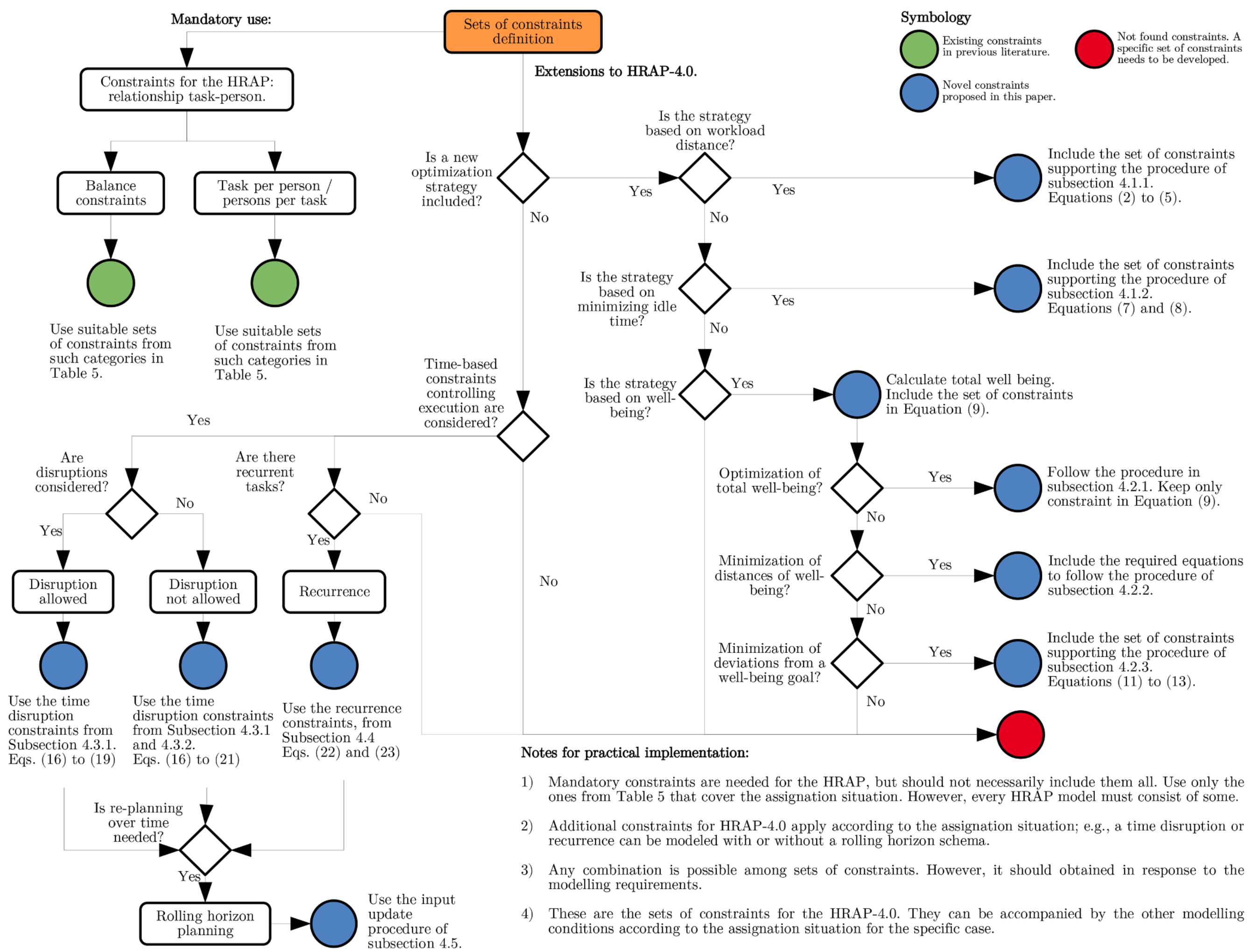


Fig. 3. Decision tree: Sets of constraints.

minimum to consider the model to be an HRAP one. Then the additional categories for workload balance, disruption, recurrence of the rolling horizon mode, and human well-being are discriminated through different paths according to the modeling situation. Notes for avoiding considerable complexity are also provided, especially during the resolution process.

3.3. Additional directions to model HRAP-4.0 with Linear Programming

Using linear programming is suggested, given the convex nature of its solution space. Therefore, Mixed Integer Linear Programming (MILP) and Multi-Objective Linear Programming (MOLP) are the best options. For example, MOLP is suggested using the combinations of the optimizations strategies in Fig. 2. Not all the equations in Tables 4 and 5 are linear representations, but they all have their linear counterpart, obtained through proper linearizations. The linearization operations are beyond the scope of this work, but readers can find further instructions in Williams (2013). All the proposals in this work for HRAP-4.0 extensions are linear.

4. Formulation extensions for the HRAP-4.0

This section aims to answer RQ4: novel formulations for HRAP-4.0. Modeling extensions are developed based on the identified building blocks categories of the optimization strategies and sets of constraints presented in Table 1. Note that these categories are included in the decision trees of the reference framework (as blue nodes in Figs. 2 and 3). The HRAP-4.0 is based on combining the current building blocks categories of Tables 4 and 5 (green nodes in Figs. 2 and 3) with those presented herein. Five proposals are presented for the building block of optimization strategies; two for the workload balance category and three for the human well-being category. Four proposals are defined in the building blocks of constraints; two for the disruption constraints category, one for the recurrence category, and an additional category for the constraints required to model the new optimization strategies whenever necessary. Different proposals are formulated for the same building block category representing alternative possibilities of HRAP-4.0, and it is necessary to select only one of them. The proposals were developed so that they covered the entire set of the previously discussed drawbacks and are summarized in Table 6.

All the proposals are developed using Mixed Integer Linear Programming (MILP). Examples are also presented to show the model's validity.

4.1. Modelling optimization strategies based on the workload balance and idle time

4.1.1. Minimization of the workload distances

The strategy minimizes the maximum distance in the total workload among all the persons instead of the distance to a predefined average or level as previously modeled. Minimizing such an objective function is equivalent to making the person's workload more similar and to,

Table 6
Extension proposals for HRAP-4.0.

Building block	Category	Proposals	Subsection
Optimization strategies	Workload balance and idle time	Minimize workload distances	4.1.1
		Minimize maximum idle time	4.1.2
	Human well-being	Optimize total well-being	4.2.1
		Minimize well-being distances	4.2.2
		Minimize deviations from a goal well-being	4.2.3
Sets of constraints	Time disruption	Allow disruption	4.3.1
		Avoid disruption	4.3.2
	Recurrence	Reccurent jobs in a plannin horizon	4.4
	Supporting	Auxilar constraints supporting optimization strategies	-
	Rolling horizon modelling	Update input data for succesive iterations of the HRAP-4.0 model	4.5

therefore, striking the workload balance. Let's consider the following decision variables:

- $\mathbf{W}^p$ Continuous non-negative variable representing the total workload assigned to person p.
- $\mathbf{C}^{pp'}$ Continuous non-negative variable representing the absolute distance between the workload of person p and the workload of any other person p'.
- $\mathbf{D}$ Continuous non-negative variable representing the maximum of the distances among all persons.

and the following set:

$\phi_{(j,p)}$ Set of tasks that can be performed by each person in the workforce.

Optimization strategy: it is done by the mathematical formulation as follows:

$$\min \Rightarrow \mathbf{D} \tag{1}$$

Subject to:

$$\mathbf{D} \geqslant \mathbf{C}^{pp'}, \quad \forall p,\ p'. \tag{2}$$

$$\mathbf{C}^{pp'} \geqslant \mathbf{W}^p - \mathbf{W}^{p'}, \quad \forall p, p' > p. \tag{3}$$

$$\mathbf{C}^{pp'} \geqslant \mathbf{W}^{p'} - \mathbf{W}^p, \quad \forall p, p' > p. \tag{4}$$

Where:

$$\mathbf{W}^p = \sum_{j \in \phi_{(j,p)}} \mathbf{t}_j^p \mathbf{X}_j^p, \quad \forall p \in \phi_{(j,p)} \tag{5}$$

Eq. (2) obliges $\mathbf{D}$ to take the maximum value of the distances among all persons. It holds that $\mathbf{D}$ could be higher than this maximum but, as the objective is to minimize it, it always aims to be lower, and the most negligible value that it can take is the maximum of distances. Eqs. (3) and (4) calculate the absolute value of the workload distance between every pair of persons (p and p'), while Eq. (5) computes the total workload assigned to each person. This strategy strikes a balance because it minimizes the maximum of distances. It is the equivalent of making all the values similar to one another. Example 1 is presented to show how this works.

Example 1. *Consider that a set of tasks with a total workload of 190 min should be assigned to five persons (p_i, $i = 1,\ldots,5$). Consider the workload values (variable $\mathbf{W}^p$) as shown in Table 7, resulting from three suitable solutions. If the average workload were maximized per person, each solution would be optimal unless the imbalance among different persons was essential (such as person 2 and 4 in solution 1). It is the main drawback of this strategy.*

Table 7
Workload values ($\mathbf{W}^p$).

Solution	Person p_1	p_2	p_3	p_4	p_5	Total (min)	Av. (min)
1	40	56	47	17	30	190	38
2	40	40	47	33	30	190	38
3	40	40	40	33	37	190	38

If the imbalance is considered, the differentiating factor among solutions would be assigning people with similar workloads. Take a look at solution 1; the upper and lower workloads are with persons 2 and 4, respectively. To make workloads similar, move 16 min (which would imply reassigning tasks) from person two to four, which results in solution 2. Persons one and two now have the same workload, and person four is closer to the other. By applying the same logic, in solution 2, move 7 min from person three to person five, which results in solution 3. There are three persons with the same workload, and the two remaining ones are close to them. Solution 3 is, thus, the best option. To validate that the new optimization strategy follows the same logic, consider the values of variable $\mathbf{C}^{pp'}$ shown in Table 8. They are all positive because they are absolute values given Eqs. (3) and (4).

Note that in each solution, the value of variable $\mathbf{D}$ is always the maximum distance given Eq. (2). Because the optimization strategy consists of minimizing $\mathbf{D}$, it always chooses the solution with lower value. It is homologous of balancing each's person workload. It is trivial to see that a global minimum is possible, where the five persons are given 38 min ($\mathbf{D} = 0$).

4.1.2. Minimization of the maximum idle time

This strategy is especially beneficial for those cases in which the set of tasks is long enough to obtain a distribution in which all the persons can be assigned near their availability time. This balance is achieved if the total idle time for everyone is minimum or at least similar. It is helpful for those cases in which people do not have the same availability time.

There are two ways to model this: the first is to use the same balance approach based on distances, as explained in the previous subsection, but by changing the total workload for an idle time percentage. The second option is to consider the following additional decision variables:

- $\mathbf{I}^p$ Continuous non-negative variable representing the idle time percentage for person p related to the total available time.
- $\mathbf{I}^*$ Continuous non-negative variable representing the maximum idle time percentage among all the persons in the workforce; this is the maximum value of the variable $\mathbf{I}^p$.

and the parameter:

$\mathbf{a}^p$ Total available time of person p on the working shift.

Optimization strategy: it consists of striking a balance via the minimization of the idle time percentage. To do so, consider the objective in Eq. (6):

$$\min \Rightarrow \mathbf{I}^* \tag{6}$$

Subject to:

$$\mathbf{I}^* \geqslant \mathbf{I}^p, \quad \forall p. \tag{7}$$

Where:

$$\mathbf{I}^p = 1 - \frac{\mathbf{W}^p}{\mathbf{a}^p}, \quad \forall p. \tag{8}$$

Eq. (7) ensures that variable $\mathbf{I}^*$ always takes the maximum value of $\mathbf{I}^p$. Finally, Eq. (8) computes the idle time fraction for every person as a fraction of their available time in the working shift.

Table 8
Distance values ($\mathbf{C}^{pp'}$).

	Solution 1					Solution 2					Solution 3				
	p1	p2	p3	p4	p5	p1	p2	p3	p4	p5	p1	p2	p3	p4	p5
p1	0	16	7	23	10	0	0	7	7	10	0	0	0	**7**	3
p2	-	0	9	**39**	26	-	0	7	7	10	-	0	0	**7**	3
p3	-	-	0	30	17	-	-	0	14	**17**	-	-	0	**7**	3
p4	-	-	-	0	13	-	-	-	0	3	-	-	-	0	4
p5	-	-	-	-	0	-	-	-	-	0	-	-	-	-	0
Max (**D**)			39					17					7		

Table 9
Idle time values ($\mathbf{I}^p$).

	Person	p1	p2	p3	p4	p5	
	$\mathbf{a}^p$	60	60	50	40	40	$\mathbf{I}^*$
$\mathbf{I}^p$	Sol. 1	33.3%	6.6%	6.0%	57.5 %	25 %	57.5 %
	Sol. 2	33.3%	33.3%	6.0%	17.5%	25.0%	33.3%
	Sol. 3	33.3%	33.3%	20.0%	17.5%	7.5%	33.3%
	Sol. 4	25.0%	25.0%	20.0%	25.0%	25.0%	25.0%

Example 2. *Consider the same case as Example 1. Assume an additional solution 4 that generates workload values of 45, 45, 40, 30, and 30 min for persons 1, 2, 3, 4, and 5, respectively. The idle time values (variable $\mathbf{I}^p$) are calculated with Eq. (8), and the $\mathbf{a}^p$ parameter values are shown in Table 9. The maximum idle time $\mathbf{I}^*$ for each solution holds due to Eq. (7) and also to the objective in Eq. (6). Readers can check that if the minimization of the distances presented in Section 4.1.1 is applied, the new solution 4 has a maximum distance of 15 min. Therefore, it does not override solution 3 as the best one in that circumstance. Now take a look at the idle time values in Table 9. Note from the previous set of solutions that solution 2 has the same maximum idle time as solution 3. In practice, both solutions seek a fair distribution, given that the fraction of the shift worked by each person tends to be the same. Therefore, misusing people in the same percentage of their shift is equally wrong regardless of the specific person (unless additional constraints are considered).*

Look at the idle time values for solution 4. Note how its maximum ($\mathbf{I}^*$) is smaller than solutions 2 and 3. This implies a better balance in the idle time per person. Even when the total workload to distribute is still the same, the average workload is also the same in all the sets of solutions (38 min). Solution 4 is, thus, that which should be chosen. As the optimization strategy consists of minimizing $\mathbf{I}^*$, it directly chooses solution 4 as required.

4.2. Modeling optimization strategies based on the occupational well-being

Occupational well-being is conceptualized as the task execution strategy while maintaining both physical and emotional propitious conditions for operators. Usually, each person has skills and preferences for some tasks than others. This can conflict with the flexible workforce concept, but it can still be considered. A high training level can narrow the productivity rates and working well-being gap but not necessarily erase it. Then one can still weigh it up. It can impact both personnel motivation and quality during execution. Besides, some risk factors can affect prioritization when selecting the tasks. To model this aspect, consider the following additional model components:

$\mathbf{s}_j^p$ parameter representing the measure of successful well-being of executing task j by person p.
$\mathbf{S}^p$ Continuos non-negative variable representing the successful well-being assigned to person p.

Remember that set $\phi_{(j,p)}$ can directly exclude specific task-person combinations because they are not feasible for some well-being condition. Parameter $\mathbf{s}_j^p$ should be expressed in two ways depending on the optimization requirement:

- **Case 1**: well-being is defined as the degree of a person's personal satisfaction while executing a task. A subjective methodology can measure operators' perceptions of their satisfaction during task execution. Minimum requirements, such as the satisfaction of putting qualifications into practice or interacting with high technology machinery (or the other CPS elements), can be weighted here.
- **Case 2:** well-being is defined as the physical risk level of performing the task. This level should be a weighted homologation of the existing ergonomic risk factors. In the CPS collaborative workstation, the more risky tasks intend to be given to machines to protect persons as much as possible. Interpret objective function Z as the result of the best task-person combination that avoids physical risks for people. The physical risk of a task is not inherent to the person but to the task itself.

The total well-being assigned to a person is given by Eq. (9).

$$\mathbf{S}^p = \sum_{j\in\phi_{(j,p)}} \mathbf{s}_j^p \mathbf{X}_j^p, \quad \forall p \in \phi_{(j,p)}. \tag{9}$$

4.2.1. Optimization of the total well-being

This strategy consists of optimizing total well-being as follows: define an objective function Z so that

$$Z = \sum_p \mathbf{S}^p. \tag{10}$$

If $\mathbf{s}_j^p$ refers to case 1, Z should be maximized, or minimized otherwise.

4.2.2. Minimization of distances of well-being

To implement this strategy, use the same methodology in Section 4.1.1 for the minimization of workload distances. To do so, define $\mathbf{W}^p = \mathbf{S}^p$ and calculate it with Eq. (9) instead of Eq. (5). Interpret $\mathbf{C}^{pp'}$ as distances of total well-being. The rest of the optimization process is applied in the same way as in Section 4.1.1.

4.2.3. Minimization of the deviations from a goal well-being

This strategy defines a distance from the total well-being assigned to a person to a predefined goal. To do so, consider the following modeling components.

$\mathbf{g}^p$ parameter representing the predefined goal of well-being for every person p.
$\mathbf{G}^p$ continuous non-negative variable representing the gap of well-being of every person p to the predefined goal.

Note that $\mathbf{g}^p$ can be defined as an average well-being per person (same value for each person), or even in particular for each one if required. Gaps should be obtained as depicted in Eq. (11)

$$\mathbf{G}^p = |\mathbf{S}^p - \mathbf{g}^p|, \quad \forall p. \tag{11}$$

Note that $\mathbf{G}^p$ is an absolute value. To represent it by linear programming, two additional constraints are required, as shown in Eqs. (12) and (13)

$$\mathbf{G}^p \geqslant \mathbf{S}^p - \mathbf{g}^p, \quad \forall p. \tag{12}$$

$$\mathbf{G}^p \geqslant \mathbf{g}^p - \mathbf{S}^p, \quad \forall p. \tag{13}$$

The optimization strategy minimizes the distances between persons of the gaps of the well-being goal. For this purpose, there are three different ways:

Optimize the average distance: to use this strategy, it is necessary to calculate the objective function as the average value as follows:

$$Z = \sum_p \frac{\mathbf{G}^p}{P}, \tag{14}$$

where P is the total quantity of persons in the workforce. Once again, if $\mathbf{s}_j^p$ is from case 1, then Z should be maximized, or minimized otherwise.

Minimize the maximum distance: to use this strategy, it is necessary to calculate the objective function as the maximum $\mathbf{G}^p$. To do so, minimize objective function Z given that

$$Z \geqslant \mathbf{G}^p. \tag{15}$$

This strategy is proper when $\mathbf{s}_j^p$ is defined from case 2.

Minimize the distances among persons: to perform this strategy, recall the minimization of distances used in Section 4.1.1. For this purpose, consider $\mathbf{W}^p = \mathbf{G}^p$ and calculate it with Eq. (11) (Eqs. (12) and (13) are also be needed). Then follow the same procedure presented there. Unlike Section 4.2.2, minimization is of the gaps from a goal for every person, and not among the distances in the total assigned well-being.

4.3. Modelling multiple time-lapse task executions with and without disruptions in a dynamic environment

In cases where some unexpected event occurs, the task can be interrupted. It requires reallocation, which considers the workload already performed and the one still pending. This fact is fundamental when task execution requires more than one time-lapse.

In a multi-time period schema, tasks can take more than one period on a planning horizon. It is common to find the term time period in the literature, which refers to days, weeks, months, or years. This work calls it *time-lapses* by understanding that a time-lapse has a predefined length that can be shorter than 1 day (e.g., 10 or 15 min). For a multi-time-lapse planning horizon, it might be needed to execute the task in consecutive lapses. However, it may also be possible to interrupt the task in one time-lapse and continue in further lapses, but not consecutive ones. To model these two cases, consider the additional model elements:

- l Index of time-lapses ($l = 1, \ldots, L$).
- $\mathbf{l}_j$ Parameter representing the remaining number of time-lapses to complete the task j. It is assumed that one task can only be allocated to one person in one same time-lapse. If task execution has not started, this parameter coincides with the total time-lapses required to complete it.
- $\mathbf{X}_{jl}^p$ Binary decision variable with a value of 1 if task j is assigned in lapse l to person p, and 0 otherwise.
- $\mathbf{A}_{jl}^p$ Binary decision variable with a value of 1 if task j is not assigned in lapse $l-1$, but assigned in lapse l (the job execution is activated in lapse l).
- $\mathbf{O}_{jl}^p$ Binary decision variable with a value of 1 if task j is assigned in lapse $l-1$, but not assigned in lapse l (the execution of the job stopped (put off) in lapse l).

As previously mentioned, it is necessary to revise the allocation of tasks to the persons in a dynamic environment. Depending on the situation, this can be done by period or event. HRAP-4.0 should be run with a fixed periodicity when replanning by period. The replanning period is usually assumed to be an integer number of time lapses. If the replanning period is shorter than $\mathbf{l}_j$, one task may start before. This situation is also possible when an event generates replanning. Indeed the online control performed in the I4.0 context allows the identification of unexpected events that can interrupt the processing of some tasks. In both cases, one task should have already started and not been finished. Therefore, the following parameter should be defined:

- $\mathbf{x}_j^p$ Binary parameter with a value of 1 if the task j assigned to the person p has already started at the beginning of the planning horizon and 0, otherwise.

4.3.1. Modelling multi time-lapse execution with disruptions

If time disruptions are allowed, it is possible for a person to execute a task partially and to continue with it in further time-lapses not consecutive. This idea is illustrated in Fig. 4.

The following sets of constraints suffice to model this situation:

$$\mathbf{X}_{jl}^p = \mathbf{x}_j^p + \mathbf{A}_{jl}^p - \mathbf{O}_{jl}^p, \quad \forall j, p, l = 1. \tag{16}$$

$$\mathbf{X}_{jl}^p = \mathbf{X}_{j(l-1)} + \mathbf{A}_{jl}^p - \mathbf{O}_{jl}^p, \quad \forall j, p, l > 1. \tag{17}$$

$$\mathbf{A}_{jl}^p + \mathbf{O}_{jl}^p \leqslant 1, \quad \forall j, p, l. \tag{18}$$

$$\sum_l \mathbf{X}_{jl}^p = \mathbf{l}_j, \quad \forall j, p. \tag{19}$$

Eq. (16) is required because of the tasks that started prior to the planning horizon beginning. The sets of constraints in Eqs. (17) and (18) ensure that a task can have multiple beginnings (disruptions). However, given the balance set of constraints in Eq. (19), the total assigned time-lapses should be precisely those required to finish it. To allow one same task to be performed by more than one person, modify Eq. (19) by including index p in the summation over the variable $\mathbf{X}_{jl}^p$ and getting rid of the definition of the Equation.

4.3.2. Modelling multi time-lapse execution without disruptions

When the disruptions are not allowed, task workload should be done in consecutive time-lapses (Fig. 5).

For this case, only one activation and deactivation time-lapse are allowed. This is done by including the constraints in Eqs. (20) and (21).

$$\sum_l \mathbf{A}_{jl}^p = 1, \quad \forall j, p. \tag{20}$$

$$\sum_l \mathbf{O}_{jl}^p = 1, \quad \forall j, p. \tag{21}$$

Example 3. *Consider the example shown in Fig. 6.*

For a task requiring five time-lapses, X_{jl}^p takes the value of 1 from lapses 3 to 7. A_{jl}^p should take a value of 1 in lapse $l = 3$, and 0 elsewhere to make the approach possible. O_{jl}^p should take a value of 1 in lapse $l = 7$, and 0 elsewhere. It is ensured by Eqs. (16)–(18). The requirement of five lapses holds because of Eq. (19). Note that A_{jl}^p and O_{jl}^p have only one value of 1 because no disruptions are allowed. This holds because of Eqs. (20) and (21). If these Equations are not considered, A_{jl}^p and O_{jl}^p can have more than one value of 1, and this allows disruptions.

In summary, the sets of constraints in Eqs. (16)–(21) are called *time disruption constrains*. Their recommended use is:

- For an assignation as illustrated in Fig. 4, use the constraints in Eqs. (16)–(18).

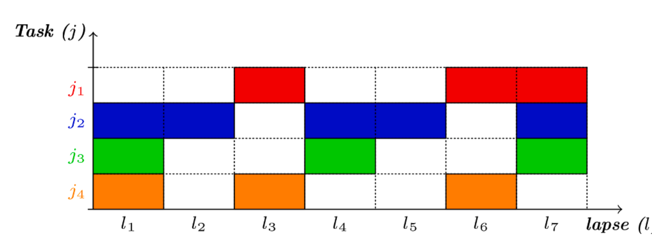


Fig. 4. Task execution with disruptions.

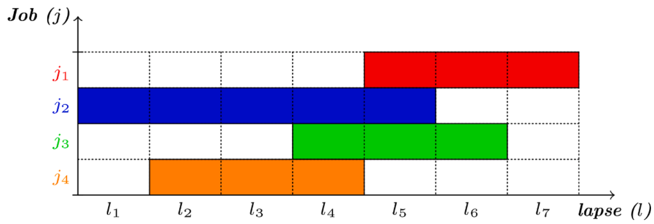


Fig. 5. Lapse assignation without disruptions.

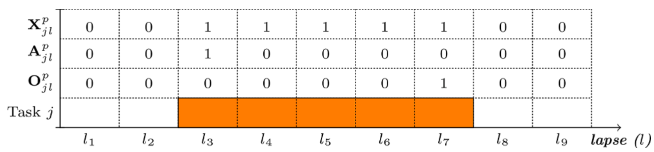


Fig. 6. Assignation through time disruption constraints.

- For an assignation as illustrated in Fig. 5, include additional constraints in Eqs. (20) and (21).
- For both previous cases, include Eq. (19).
- For the particular case which requires the possibility of several people executing a task, the same formulas can be applied, but summing up on index p. It is useful in, for example, those cases in which one person starts a task and another finishes it.

4.4. Modelling recurrence in task executions on a planning horizon

Recurrence is inherent to those tasks that need to be executed more than once on a planning horizon. For example, manual cash accounting in a supermarket service station or preventive manual machine calibration in a workstation in manufacturing. Both can be controlled and monitored by a CPS, and HRAP-4.0 can deliver their assignation. Thus recurrent tasks should be executed in every determined time window on a planning horizon called $\mathbf{L}$ composed of several time-lapses. This situation applies mainly to those cases in which no disruptions in execution are allowed. If a disruption is allowed, and unless the subsequent execution can also be replanned, both can overlap. Thus performing/finishing two cash accounts or machine calibrations simultaneously, or even too close to one another, is unsuitable. Because of this, keeping recurrence modeling for the non-disruption case is suggested.

To implement task recurrence for HRAP-4.0, consider the following additional modeling components:

- e — Index referring to the number of the task executions on planning horizon $\mathbf{L}$ (one task can have more than one execution on the planning horizon).
- $\xi_{(e,j)}$ — Set indicating the number of executions in which each task should be performed. This set contemplates task recurrence.
- $\mathbf{m}_j$ — Integer parameter representing the number of time lapses required between consecutive executions for each recurrent task j.

Then modify the parameter.

- $\mathbf{l}_j^e$ — Parameter representing the remaining number of time-lapses to complete task j during execution e. It is assumed that one task can only be allocated to one person in the same time-lapse. If task execution has not started, this parameter coincides with the total time-lapses required to complete it.

The variables presented in Section 4.3 need to be modified as follows:

- $\mathbf{X}_{jl}^{pe}$ — Binary variable with a value of 1 if task j is assigned to person p, for execution e in lapse l, and 0 otherwise.
- $\mathbf{A}_{jl}^{pe}$ — Binary decision variable with a value of 1 if task j is not assigned in lapse $l-1$ for execution e and assigned in lapse l (the execution e of the task is activated in lapse l).
- $\mathbf{O}_{jl}^{pe}$ — Binary decision variable with a value of 1 if task j is assigned in lapse $l-1$ for execution e and not assigned in lapse l (the execution e of the task is put off in lapse l).

Eq. (19) needs to be extended as follows:

$$\sum_{l}\mathbf{X}_{jl}^{pe} = \mathbf{l}_j^e, \qquad \forall (e,j) \in \xi_{(e,j)}, p. \tag{22}$$

The set of constraints in Eq. (23) is introduced to ensure that the quantity of lapses between successive executions for one same task is maintained. This is known as the *recurrence constraint*.

$$(l+\mathbf{m}_j)\mathbf{X}_{jl}^{pe} \leqslant \mathbf{M}(1-\mathbf{X}_{jl'}^{pe'}) + l'\cdot\mathbf{X}_{jl'}^{pe'}, \qquad \forall (e,j) \in \xi_{(e,j)}, l, e' > e, l' > l, p. \tag{23}$$

Note that, if it was needed to model recurrence while maintaining the rules for the time disruptions set out in Section 4.3 in some cases, all the disruption constraints remain the same, but include the new index e and set $\xi_{(e,j)}$ (the same extension as in Eq. (22)). This should lead to a result in which the time window between successive executions remains (the last time-lapse of execution e should always be at least $\mathbf{m}_j$ lapses before the first lapse of execution $e+1$, no matter what the number of disruptions in between). By doing this, modelers might have to check/modify set $\xi_{(e,j)}$ if the previous number of executions to be made is no longer suitable.

4.5. HRAP-4.0 in a rolling horizon mode

As previously indicated, sudden or controlled events can lead to the need to reorganize successively (replanning) task assignation on a planning horizon in line with the information collected and provided by the HCPS. Replanning supports reflecting on the model of the proper monitoring of task execution in a dynamic environment and because of this, implementing HRAP-4.0 in a rolling horizon mode is necessary. As stated before, when replanning by period, HRAP-4.0 should be run with fixed periodicity, named the replanning period. When replanning by event, HRAP-4.0 should be executed every time the unexpected or planned event occurs. This subsection describes the procedure for replanning by period by understanding that replanning by event requires the same model updates, but an event occurs every time. The rolling horizon can be implemented for HRAP-4.0 by including all the novel characteristics proposed in this work, i.e., all the optimization strategies and additional constraints to model disruptions and recurrence. To fulfill this purpose, consider the following additional model components:

- i — Index of iteration in the rolling horizon mode. One iteration corresponds to one replanning. For replanning by period, it should be an integer multiple of time-lapses.
- E_j — Integer parameter representing the number of executions for each task j in the current iteration i. It is the maximum value of e for each task in the set $\xi_{(e,j)}$.
- $\mathbf{n}_i$ — Integer parameter representing the number of time-lapses in iteration i.

Consider the case of two successive iterations in the model. At the end of the first iteration i, the execution of a task might be ongoing. So at the beginning of the following iteration $i+1$, execution still continues. Further executions of that same task in iteration $i+1$ will depend on the planning horizon length. There are two possibilities:

1) **Fixed-length planning horizon**: this is when the same planning horizon length should be maintained throughout iterations. For the new iteration $i+1$, $\mathbf{n}_i$ lapses have already been consumed. The exact quantity of the new time-lapses should be added at the end of the horizon to maintain the original length. Lapse numbers should be reorganized to always start in lapse 1 (lapse l in iteration 1 becomes lapse $l-\mathbf{n}_i$ in iteration $i+1$). This same logic applies to further iterations. Note that, for this rolling horizon case, the number of possible iterations is infinite. The total quantity of iterations to be performed depends on planners' needs.

2) **Decreasing-length planning horizon**: this is when length decreases throughout iterations. For the new iteration $i+1$, the new length of the planning horizon is $\mathbf{L} - \mathbf{n}_i$ lapses because none is added at the end. Thus length starts decreasing with each successive iteration. The quantity of iterations is, in this case, finite.

For each iteration in the rolling horizon, an update of input parameters $\mathbf{x}_j^p, \mathbf{l}_j^e, E_j$ and set $\xi_{(e,j)}$ is required. This update takes place outside the model, and it only needs calculation for those tasks being performed at the end of the current iteration i. The update for parameter $\mathbf{x}_j^p$ before iteration $i+1$ is done with Eq. (24).

$$\mathbf{x}_j^p = \begin{cases} 1 & \text{, if task } j \text{ is being performed at the end of iteration } i, \\ 0 & \text{, otherwise.} \end{cases} \tag{24}$$

For the parameter $\mathbf{l}_j^e$, the execution of the task being performed at the end of iteration i becomes the first in the next iteration $i+1$. The $\mathbf{l}_j^e$ value should be updated as follows:

$$\mathbf{l}(i+1)_j^e = \begin{cases} |\mathbf{n}_i - \mathbf{l}(i)_j^e| & , \quad \forall j, e = 1. \\ \mathbf{l}(i)_j^e & , \quad \forall j, e > 1. \end{cases} \tag{25}$$

Note that factor $|\mathbf{n}_i - \mathbf{l}(i)_j^e|$ is expressed as an absolute value given the possibility of a case in which the remaining period to complete a task is longer than the time window for successive iterations. If such a case is not possible given the modeling situation, there is no need to express this factor as an absolute value.

Defining the maximum e value is still necessary to consider that it directly affects $\mathbf{l}(i)_j^e$ in iteration $i+1$. Note that the previous value might not be reached for parameter E_j depending on the new planning horizon length $\mathbf{L}$. If the planning horizon is long enough, the E_j value should be maintained. In general, if the planning horizon length (according to the two described cases) is suitable for maintaining the E_j value, then set $\xi_{(e,j)}$ remains the same. On the contrary, there are two possibilities; one is that the planning horizon length does not permit the same E_j value to remain, and it needs to be lowered. In the second case, the planning horizon allows more executions of the same task, and E_j increases. In both cases, set $\xi_{(e,j)}$ needs to be updated with the executions' exclusion/inclusion, respectively, and parameter $\mathbf{l}_j^e$ should reflect the same quantity of executions for $e > 1$ in Eq. (25).

Example 4. Consider the following situation:

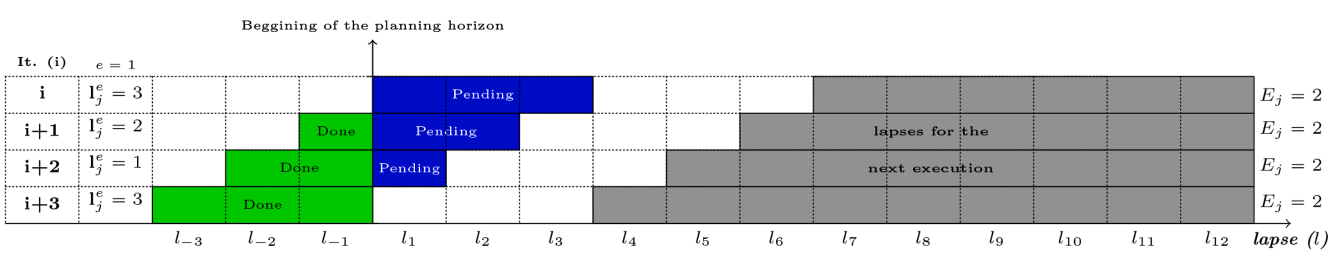
- A planning horizon comprises 12 time-lapses, $\mathbf{L} = 12$.
- A recurrent task requires 3 time-lapses ($\mathbf{l}_j^e = 3$). It has been assigned for execution from lapse 1 to 3.
- At least 3 lapses between successive executions are required ($\mathbf{m}_j = 3$).
- The task should be executed twice ($E_j = 2$) on planning horizon $\mathbf{L}$.
- For this example, the model should be executed in four successive iterations, $i \in \{1, .., 4\}$. Note that each iteration ocurrs with every time-lapse consumption ($\mathbf{n}_i = 1$), which may vary depending on the specific situation.

The assignation of such a recurrent task in a rolling horizon mode is illustrated in Fig. 7. Consider in iteration $i+1$ that one lapse has already gone. Then, based on the previously described planning horizon length cases, there are two ways for the second task execution in further iterations:

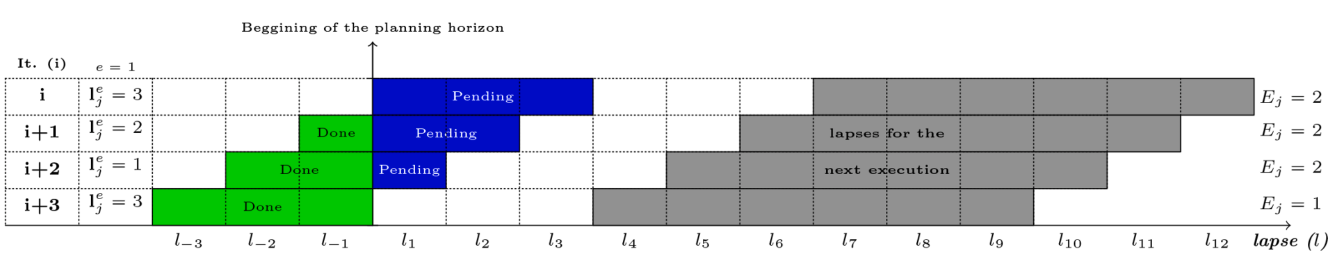
(a) **Fixed-length (Fig. 7a):**
Note that for iteration i, the second execution can happen in lapses 7–12. For iteration $i+1$, the second execution is opened to be replanned from lapse 6 to lapse 12. The same applies to iterations $i+2$ and $i+3$. Note for iteration $i+3$ that the first task execution is finished. For this case, the holding period of three lapses between successive executions is still required. In this scenario of Fig. 7a, parameter E_j can be maintained and, thus, set $\xi_{(e,j)}$ always remains the same.

(b) **Decreasing-length (Fig. 7b):**
For iteration $i+1$, the new planning horizon length is $12 - 1 = 11$ lapses. The second execution is open to be replanned for lapses 6 to 11. As the planning horizon length decreases, note in iteration $i+3$ that parameter E_j can no longer be maintained and should be decreased. For this case, set $\xi_{(e,j)}$ needs to be modified, subtracting one task execution.

Note in both cases that the time window between successive executions holds due to Eq. (23). The update for parameter $\mathbf{l}_j^e$ holds due to Eq. (25).



(a) Fixed-length planning horizon.



(b) Decreasing-length planning horizon.

Fig. 7. Recurrent tasks in a rolling horizon schema.

5. Methodological guidelines for implementation

At this point, it is necessary to integrate and use the reference framework and its building blocks in real I4.0 contexts. Methodological guidelines for implementing HRAP-4.0 in a CPS environment are proposed that also deal with RQ5. It is important to note that the construction of the entire CPS is beyond the scope of this work. Readers are referred to Lee et al. (2015) to find more information about this. The proposed procedure comprises eight main steps (Fig. 8):

1. Analyze the assignation situation: identify the type of tasks to be assigned, their execution time, conditions, possible specific qualifications for the flexible workforce, planning horizon, resources availability, etc. Specific tasks might be needed for the integration

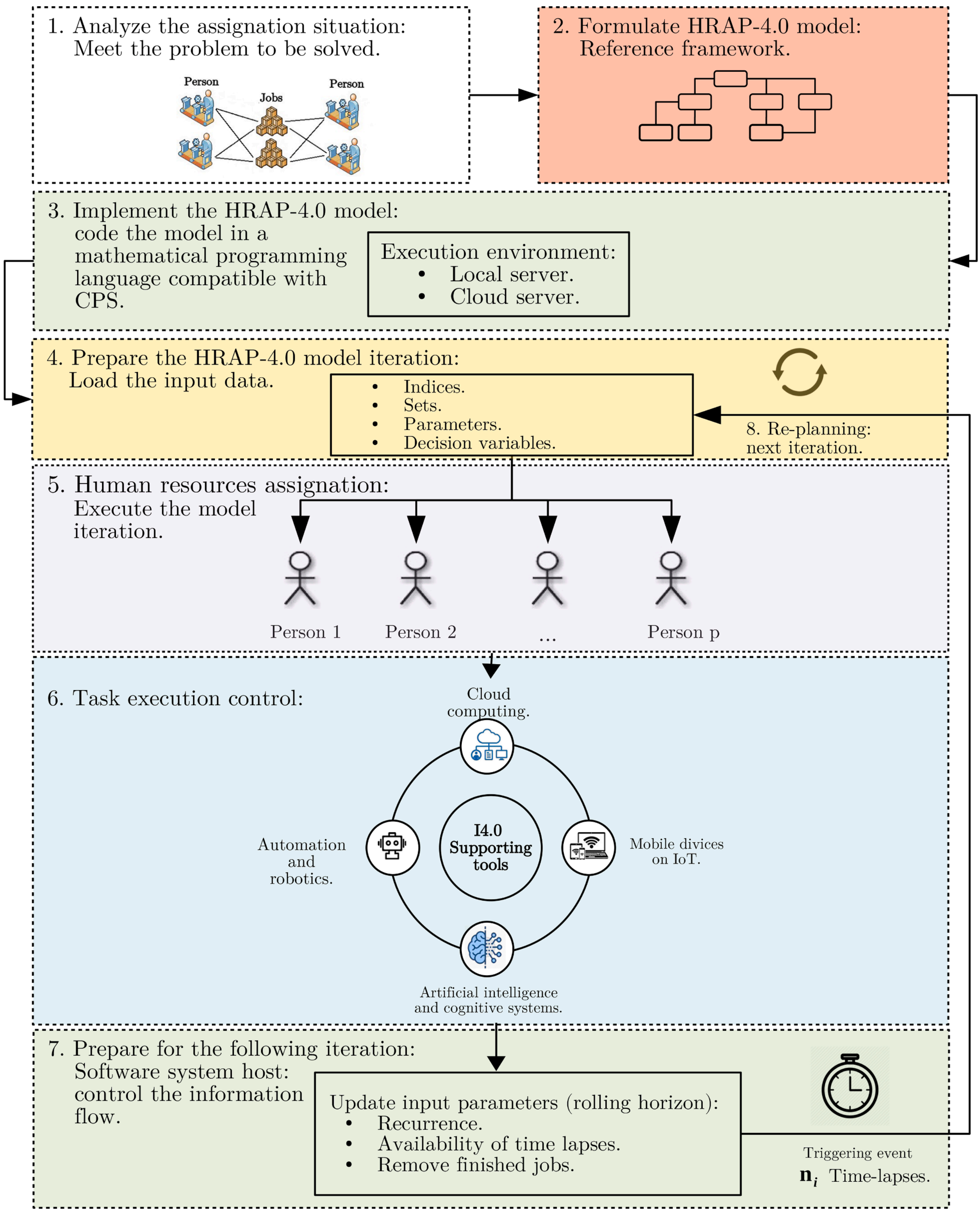


Fig. 8. Methodology for implementing HRAP-4.0 in the frame of a CPS.

with CPS. For example, the reporting of advanced task execution, manual data entry, manipulation of actuators, among others. The I4.0 requirements of Table 1 can be helpful at this point.
2. Formulate the HRAP-4.0 model: At this point, the reference framework should be used to model the HRAP-4.0 situation in hand defined in step 1.
3. Implement the HRAP-4.0 model: code the model with a mathematical modeling language to translate it into a readable machine form (e.g., MPL software), and choose the solver (e.g., Gurobi). This step also includes arranging the model input parameters and datasets required. Verifying the compatibility between the language and the software system (s) running in CPS for this step is strongly recommended.
4. Prepare the HRAP-4.0 model iteration: it requires loading the input data for the iteration. If the model is executed for the first time (first iteration), provide it with the initial information of the parameters and datasets. If the iteration is not the first one, the model should be fed the data updated from CPS through the IoT.
5. Human resources assignation: solve the model with the selected solver and the updated input data. The iteration result consists of the human assignation arrangement. This solution should be delivered to CPS. CPS facilitates the assignation arrangement being transmitted to the workforce.
6. Task execution control: the workforce starts performing the tasks assigned for the current period in the previous step. They interact with CPS to provide information in return about the progress of each task execution. CPS constantly monitors their progress. If the progress of tasks is as planned, no new iteration of the HRAP-4.0 model is needed until the replanning period elapses. Otherwise, a triggering event might provoke the new iteration to re-assignation that adapts to environmental changes. Therefore, this module provides real-time information that can trigger the following activity (step 7) if an event occurs before the replanning period elapses. Besides, the information captured by CPS can be used to measure operators' performance and compare it to other assignations. Operators' performance information can be beneficial for defining training programs and updating input parameters of operators' efficiency to perform better future assignments.
7. Prepare for the following iteration: with a rolling horizon schema, performing the parameters update described in Section 4.5 is required. For this purpose, the status reported in step 6 is needed.
8. Replanning: it consists of going to the next iteration when either the replanning period elapses or a triggering event occurs. If task execution ends in the previous iteration, this task should be removed for the next one. If a task starts or is underway in the previous iteration but is not finished, the remaining work should be updated in the following iteration. The latter is also related to task recurrence. If a recurrent task comes into play, new task executions are needed if the new planning horizon allows it. For no recurrent tasks, or even new tasks in the following iteration, there are two possibilities:
   i. If the first assignation arrangement obtained in the first model iteration is to be maintained (the same tasks for the same persons), then the HRAP-4.0 model later acts as a re-assignation tool. The already assigned tasks for the person can be reassigned in the upcoming time lapses on the planning horizon. To make this possible, the model needs a new set of tasks preassigned to each person in every model iteration.
   ii. If the first assignation arrangement obtained in the first model iteration is not maintained (the whole set of tasks can be reassigned to different persons) for the following iterations, the tasks already done and those underways are removed. The remaining tasks can be ultimately reassigned. This implies that a new assignation arrangement is obtained for every person in every iteration. After the model iteration, it goes back to the fourth step of this procedure in both cases. Then, the results of the assignation arrangement are uploaded again to the CPS to continue with the next iteration, and so on.

6. Discussion

6.1. HRAP vs. HRAP-4.0: comparison analysis

The literature shows that the HRAP is an extensively covered topic (Pentico, 2007; Bouajaja and Dridi, 2017) and remains active because of the continuous need to manage people. Therefore, this work focuses on analyzing the HRAP on the core, i.e., the specific building blocks categories that make it a way to model task allocation to labor independently of the situation, industrial sector, or the particular characteristics of the case.

Furthermore, the constant change in working conditions sustains the HRAP. This work highlights that I4.0 needs to be considered a current source of modifying factors for managing people from the classic production system (Lu, 2017) to CPS. In light of this, an HRAP version that considers these I4.0 factors becomes a clear need. This work presents the main I4.0 modifying factors in the HRAP and introduces HRAP-4.0 as an extension that generalizes it in a modular and reusable way.

For this matter, the HRAP models from the literature review are grouped into six main categories based on the purpose of their optimization strategies; (i) profit; ii) cost; iii) execution time; iv) ratios; v) permutations; vi) persons' qualifications); and into three main categories based on the nature of their constraints; (i) task per person; ii) persons per task; iii) balance constraints of assigned tasks. The specific references dealing with each category in the revised literature can be consulted in Tables 4 and 5. As readers can check, the I4.0 requirements are not reflected in the current categories of existing research works.

To bridge this gap, different optimization strategies and constraint categories are defined (Table 1). This work presents the formulation of each one. A reference framework is proposed to support the development and execution of new HRAP-4.0 models from a methodological point of view. The framework generalizes how to build HRAP-4.0 models, which is the primary goal of this work.

To model HRAP situations in the I4.0 context, it is necessary to combine already formulated categories in the literature with the new ones of this paper. Therefore, coherently defining a homogeneous nomenclature and reformulating existing models are required. Besides, when analyzing existing HRAP models and formulating new HRAP-4.0 ones, modules are identified to be reused in other HRAP-4.0 situations by employing the reference framework. Any existing formulation of Tables 4 and 5 coincides with the I4.0 proposals for the categories in Table 6 (modeled in Section 4), which are a novelty in this paper.

6.2. Research contributions

As presented in the Introduction, the contributions of this research work are manifold:

1. The main I4.0 modifying factors incorporated into HRAP-4.0 are summarized by considering the human–machine integration and the human role in CPS, process control, and information handling. This work systematically relates these factors to new building blocks to model HRAP-4.0 that generalizes the HRAP in I4.0 contexts.
2. Standard mathematical nomenclature is defined to reformulate existing HRAP models in a modular way that can be reused by the reference framework and compared to the new HRAP-4.0 formulations of this paper.
3. The new and original building blocks categories are formulated to cover I4.0 modifying factors in the HRAP-4.0. None of them is identified in the literature as part of the HRAP, which establishes its originality.
4. A reference framework is proposed to take advantage of the combination of existing and new models when facing an HRAP in a specific

I4.0 situation. This reference framework is presented in two decision trees that help modelers select and build HRAP-4.0 optimization strategies (objectives) and constraints.

5. Methodological guidelines for formulating, implementing, using, and updating the HRAP-4.0 model in a dynamic environment are proposed.

6.3. Managerial insights

This research work provides a powerful generic tool kit that allows the HRAP to be adapted in I4.0 contexts independently of sectors from the practical viewpoint. Due to non-stop technology progress, the generation of I4.0 modifying factors over the HRAP-4.0 is not expected to be static in time. It can also be generalized for newer factors on a reasonable mid-term horizon. Indeed the entire set of changes that the I4.0 has brought about to help understand production systems is still ongoing. This work presents HRAP-4.0 as a suitable way to manage people in this context. It introduces it within the frame of CPS as the basic unit of I4.0. The methodological guidelines for building, implementing, and using HRAP-4.0 models are also presented in the sequence of steps described in Section 5. From the managerial perspective, the entire set of steps is challenging, primarily when tasks are assigned to operators whose purpose relates to the interaction with CPS.

Considering that the CPS design is customized according to production system features, such tasks/jobs can differ considerably from one industrial sector to another or from one business size to the next. It is also noteworthy that step 3 of the methodological guidelines (Fig. 8), where implementing the HRAP-4.0 model, presents the challenging issue of integrating it into CPS. This issue is related mainly to software systems' compatibility. It could become a disadvantage to achieve interoperability if non-compatibilities come out. The second managerial aspect appears to be delivering and monitoring the execution of the tasks assigned by the HRAP-4.0 model (step 6 of the methodological guidelines). This control and monitoring activity should also be seen as a CPS design requirement. It can directly affect human performance, process control, and even human well-being. It should be remembered that the people in this step will closely interact with I4.0 terminals, which form part of the human–machine integration. Therefore, this approach assumes considering humans part of CPS.

The main disadvantage here is a business' access to different CPS possibilities. Depending on the options, some CPS can be very demanding for human flexibility and knowledge. That is why the human well-being trade-off is a required factor that must be considered. CPS capabilities are expected to be strongly affected by their cost. A stronger CPS, capable of considerably increasing productivity rates, can be very expensive in industrial terms. Nevertheless, the development of I4.0 tools has become progressively more accessible in cost terms thanks to newer technologies capable of doing more things at a better price. A high-performing CPS can be very demanding for people (because of workload, physical efforts, number of executions, etc.).

Finally, it is also important to stress that applying the methodological guidelines for implementation, the reference framework for HRAP-4.0, and the integration into CPS will be critically associated with the capability to execute HRAP-4.0 in a rolling horizon scheme, especially for replanning by events. The human role in CPS is significantly affected by different sources of uncertainty that demand robust re-assignations and replanning possibilities. For labor, it is hard to achieve better performance rates when execution conditions are dynamic in time. This is why replanning becomes necessary and should be appropriately considered to gain better results with HRAP-4.0, steps 7 and 8 in Fig. 8.

7. Conclusions

The HRAP faces new challenges due to the fourth industrial revolution known as Industry 4.0 (I4.0). This paper identifies the main modifying factors from I4.0 that can directly affect the HRAP. The required extension of existing HRAP models includes the derivated requirements from such a set of factors. The result is included in this paper as HRAP-4.0. The new characteristics are translated into additional building blocks of optimization strategies and constraints using MILP models. A reference framework integrated by one decision tree for optimization strategies and one for constraints is proposed to support the modeling of HRAP-4.0. These decision trees reunite previous HRAP models and the new ones defined in the I4.0 context in this paper. Finally, a methodology is presented that implements HRAP-4.0 into a CPS in a dynamic environment.

Compared to previous research, this paper contributes to different areas: 1) models formulation: developing new mathematical programming models to reflect the requirements of I4.0 factors in a modular and reusable (building blocks) way jointly with revised HRAP models; 2) Conceptual proposals for building HRAP-4.0 models using existing and new building blocks. Indeed the importance of the reference framework in this paper lies in the fact that it allows/guides modelers through the HRAP-4.0 building process. Furthermore, general directions in methodology are also provided for CPS designers and HRAP-4.0 modelers with the practical implementation. Both proposals bridge gaps identified and are contributions of this paper.

HRAP-4.0 can be seen as the new tool for assigning tasks to people in the modern production system. Its strategical development relies on the human factor being considered part of the CPS design in I4.0, HCPS. It primarily implies fairer working conditions for people in either physical or mental terms. It also permits additional task characteristics and more flexible conditions for assignment to be considered in time terms. The potential that HRAP-4.0 brings to production systems should be seen as an option in service to help people achieve better efficiency and productivity rates. It is critically important to have highly trained people to interact with machines and CPS. Such a workforce is known as a flexible workforce and is especially useful in I4.0. People are expected to have high technical levels and flexibility to react to quick changes in the production systems' reconfiguration to support traceability, flexibility, and resilience. HRAP-4.0 is imperative in response to customers' high customization/diversity and short delivery time requirements in the modern industry. The interoperability between the model's input and output data with CPS comes over as a crucial aspect.

Research limitations and further implications are outlined. The basis for developing HRAP-4.0 is the I4.0 factors identified in Table 1. More I4.0 factors might not already be considered or arise with newer technologies requiring new building blocks. Using the proposed notation and formulating it in a modular and reusable way, they should be consistently introduced into the reference framework, which should be mandatorily updated. Another limitation is that this research focuses on the mathematical programming formulation of HRAP-4.0 models but not on the solution method. Depending on the model's complexity and the available time for reassigning persons, an exact, heuristic, or metaheuristic method would be advisable for its solution. The exact option is more time-consuming than the others but, in return, provides the best solution. The time available to run the model and implement its results may depend on whether replanning is by periods or events. Usually, replanning by events requires faster solutions to increase the system's resilience and shorten recovery time. Therefore, future research lines should define which solution methods are more appropriate in each case to fully implement HRAP-4.0 models.

Finally, all the parameters of the HRAP-4.0 models are considered deterministic. However, most should be uncertain and bring about specific variability in task execution. Moreover, CPS interfaces should consider that human data are stochastic due to changes in human behavior, characteristics, and skills, which require adaptive responses. Therefore, future research directions should pay attention to how CPS can support online information collection with a multimodal interaction for data collection and data presentation to human actors. This information can be later analyzed with statistical methods to determine the input data's probability distributions to be incorporated into HRAP-4.0

models. Besides, AI can allow knowledge from these data to be drawn. Furthermore, exploring which type of knowledge is to be generated by the input and output data of CPS seems a promising research line.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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